

ANSIBLE

Assignment 33 – Onboarding- Workstation Setup, Standards & AI (Go beyond)

1. Environment Setup & Ansible Installation

Application to Use:

- **Terminal (Linux/macOS) or PowerShell (Windows with WSL/Ubuntu)**

Steps:

Create project folder and navigate into it

`mkdir -p ~/ansible-onboarding && cd ~/ansible-onboarding`

```
ubuntu@ip-172-31-23-116:~$ mkdir -p ~/ansible-onboarding && cd ~/ansible-onboarding
ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

`sudo apt update`

```
Sudo apt install python3.12-venv
```

AWS

Q Search

[Alt+S]

United States (N. Virginia)Account ID: 3111-4154-2113Adrina Colaco

```
ubuntu@ip-172-31-23-116:~/ansible-onboarding$ sudo apt install python3.12-venv
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 python3-pip-whl python3-setuptools-whl python3.12 python3.12-minimal
Suggested packages:
  python3.12-doc binutils binfmt-support
The following NEW packages will be installed:
  python3-pip-whl python3-setuptools-whl python3.12-venv
The following packages will be upgraded:
  libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 python3.12 python3.12-minimal
5 upgraded, 3 newly installed, 0 to remove and 50 not upgraded.
Need to get 10.7 MB of archives.
After this operation, 2781 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3.12t64 amd64 3.12.3-1ubuntu0.8 [2355 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3.12 amd64 3.12.3-1ubuntu0.8 [651 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3.12-stdlib amd64 3.12.3-1ubuntu0.8 [2068 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3.12-minimal amd64 3.12.3-1ubuntu0.8 [2334 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3.12-minimal amd64 3.12.3-1ubuntu0.8 [836 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-pip-whl all 24.0+dfsg-1ubuntu1.3 [1707 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-setuptools-whl all 68.1.2-2ubuntu1.2 [716 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3.12-venv amd64 3.12.3-1ubuntu0.8 [5678 B]
Fetched 10.7 MB in 0s (64.7 MB/s)
(Reading database ... 71713 files and directories currently installed.)
Preparing to unpack .../0-libpython3.12t64_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking libpython3.12t64:amd64 (3.12.3-1ubuntu0.8) over (3.12.3-1ubuntu0.7) ...
Preparing to unpack .../1-python3.12_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking python3.12 (3.12.3-1ubuntu0.8) over (3.12.3-1ubuntu0.7) ...
Preparing to unpack .../2-libpython3.12-stdlib_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking libpython3.12-stdlib:amd64 (3.12.3-1ubuntu0.8) over (3.12.3-1ubuntu0.7) ...
Preparing to unpack .../3-python3.12-minimal_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking python3.12-minimal (3.12.3-1ubuntu0.8) over (3.12.3-1ubuntu0.7) ...
Preparing to unpack .../4-libpython3.12-minimal_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking libpython3.12-minimal:amd64 (3.12.3-1ubuntu0.8) over (3.12.3-1ubuntu0.7) ...
Selecting previously unselected package python3-pip-whl.
Preparing to unpack .../5-python3-pip-whl_24.0+dfsg-1ubuntu1.3_all.deb ...
Unpacking python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Selecting previously unselected package python3-setuptools-whl.
Preparing to unpack .../6-python3-setuptools-whl_68.1.2-2ubuntu1.2_all.deb ...
Unpacking python3-setuptools-whl (68.1.2-2ubuntu1.2) ...
Selecting previously unselected package python3.12-venv.
Preparing to unpack .../7-python3.12-venv_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking python3.12-venv (3.12.3-1ubuntu0.8) ...
Setting up python3-setuptools-whl (68.1.2-2ubuntu1.2) ...
Setting up python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Setting up libpython3.12-minimal:amd64 (3.12.3-1ubuntu0.8) ...
Setting up python3.12-minimal (3.12.3-1ubuntu0.8) ...
Setting up libpython3.12-stdlib:amd64 (3.12.3-1ubuntu0.8) ...
Setting up python3.12 (3.12.3-1ubuntu0.8) ...
Setting up libpython3.12t64:amd64 (3.12.3-1ubuntu0.8) ...
Setting up python3.12-venv (3.12.3-1ubuntu0.8) ...
Processing triggers for systemd (255.4-1ubuntu8.10) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.5) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
systemctl restart networkd-dispatcher.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

CloudShellFeedback

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python3 -m venv .venv

```
ubuntu@ip-172-31-23-116:~/ansible-onboarding$ sudo python3 -m venv .venv
ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

. .venv/bin/activate

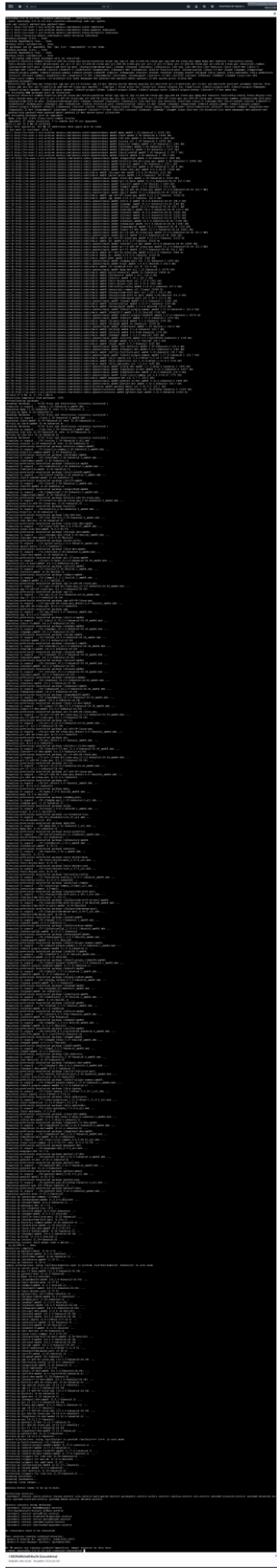
```
ubuntu@ip-172-31-23-116:~/ansible-onboarding$ . .venv/bin/activate  
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

sudo apt update

sudo apt install python3-pip python3-venv



Upgrade pip

python -m pip install --upgrade pip

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ python -m pip install --upgrade pip
Requirement already satisfied: pip in ./venv/lib/python3.12/site-packages (24.0)
Collecting pip
  Downloading pip-25.3-py3-none-any.whl.metadata (4.7 kB)
  Downloading pip-25.3-py3-none-any.whl (1.8 MB)
  1.8/1.8 MB 26.6 MB/s eta 0:00:00
Installing collected packages: pip
Attempting uninstall: pip
  Found existing installation: pip 24.0
  Uninstalling pip-24.0:
    Successfully uninstalled pip-24.0
  WARNING: Failed to remove contents in a temporary directory '/home/ubuntu/ansible-onboarding/.venv/lib/python3.12/site-packages/~ip-24.0.dist-info'.
  You can safely remove it manually.
  WARNING: Failed to remove contents in a temporary directory '/home/ubuntu/ansible-onboarding/.venv/lib/python3.12/site-packages/~ip'.
  You can safely remove it manually.
Successfully installed pip-25.3
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Install Ansible and linting tools

python -m pip install ansible ansible-lint yamllint

```
(venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ python -m pip install ansible ansible-lint yamllint
WARNING: Ignoring invalid distribution -ip (/home/ubuntu/ansible-onboarding/.venv/lib/python3.12/site-packages)
WARNING: Ignoring invalid distribution -ip (/home/ubuntu/ansible-onboarding/.venv/lib/python3.12/site-packages)
Collecting ansible
  Downloading ansible-12.1.0-py3-none-any.whl.metadata (8.0 kB)
Collecting ansible-lint
  Downloading ansible-lint-25.9.2-py3-none-any.whl.metadata (6.3 kB)
Collecting yamllint
  Downloading yamllint-1.37.1-py3-none-any.whl.metadata (4.3 kB)
Collecting ansible-core==2.19.3 (from ansible)
  Downloading ansible-core-2.19.3-py3-none-any.whl.metadata (7.7 kB)
Collecting Jinja2>=3.1.0 (from ansible-core==2.19.3->ansible)
  Downloading Jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting PyYAML>=5.1 (from ansible-core==2.19.3->ansible)
  Downloading PyYAML-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.4 kB)
Collecting cryptography (from ansible-core==2.19.3->ansible)
  Downloading cryptography-46.0.3-cp311-abi3-manylinux_2_34_x86_64.whl.metadata (5.7 kB)
Collecting packaging (from ansible-core==2.19.3->ansible)
  Downloading packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting resolvelib<2.0.0,>=0.5.3 (from ansible-core==2.19.3->ansible)
  Downloading resolvelib-1.2.1-py3-none-any.whl.metadata (3.7 kB)
Collecting ansible-compat>=25.8.2 (from ansible-lint)
  Downloading ansible-compat-25.8.2-py3-none-any.whl.metadata (3.4 kB)
Collecting black>=24.3.0 (from ansible-lint)
  Downloading black-25.9.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.manylinux_2_28_x86_64.whl.metadata (83 kB)
Collecting cffi>=1.17.1 (from ansible-lint)
  Downloading cffi-1.17.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (2.6 kB)
Collecting distro>=1.9.0 (from ansible-lint)
  Downloading distro-1.9.0-py3-none-any.whl.metadata (6.8 kB)
Collecting filelock>=3.8.2 (from ansible-lint)
  Downloading filelock-3.20.0-py3-none-any.whl.metadata (2.1 kB)
Collecting importlib-metadata>=8.7.0 (from ansible-lint)
  Downloading importlib-metadata-8.7.0-py3-none-any.whl.metadata (4.0 kB)
Collecting jsonschema>=4.10.0 (from ansible-lint)
  Downloading jsonschema-4.25.1-py3-none-any.whl.metadata (7.6 kB)
Collecting pathspec>=0.10.3 (from ansible-lint)
  Downloading pathspec-0.12.1-py3-none-any.whl.metadata (21 kB)
Collecting referencing>=0.36.2 (from ansible-lint)
  Downloading referencing-0.37.0-py3-none-any.whl.metadata (2.8 kB)
Collecting ruamel-yaml>=0.18.11 (from ansible-lint)
  Downloading ruamel-yaml-0.18.16-py3-none-any.whl.metadata (1.1 kB)
Collecting ruamel-yaml-clib>=0.2.12 (from ansible-lint)
  Downloading ruamel-yaml-clib-0.2.14-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.0 kB)
Collecting subprocess-tee>=0.4.1 (from ansible-lint)
  Downloading subprocess-tee-0.4.2-py3-none-any.whl.metadata (3.3 kB)
Collecting wcmatch>=8.5.0 (from ansible-lint)
  Downloading wcmatch-10.1-py3-none-any.whl.metadata (5.1 kB)
Collecting click>=8.0.0 (from black==24.3.0->ansible-lint)
  Downloading click-8.3.0-py3-none-any.whl.metadata (2.6 kB)
Collecting mypy-extensions>=0.4.3 (from black==24.3.0->ansible-lint)
  Downloading mypy_extensions-1.1.0-py3-none-any.whl.metadata (1.1 kB)
Collecting platformdirs>=2 (from black==24.3.0->ansible-lint)
  Downloading platformdirs-4.5.0-py3-none-any.whl.metadata (12 kB)
Collecting pytokens>=0.1.10 (from black==24.3.0->ansible-lint)
  Downloading pytokens-0.2.0-py3-none-any.whl.metadata (2.0 kB)
Collecting pycparser (from cffi==1.17.1->ansible-lint)
  Downloading pycparser-2.23-py3-none-any.whl.metadata (993 bytes)
Collecting zipp>=3.20 (from importlib-metadata==8.7.0->ansible-lint)
  Downloading zipp-3.23.0-py3-none-any.whl.metadata (3.6 kB)
Collecting MarkupSafe>=2.0 (from Jinja2>=3.1.0->ansible-core==2.19.3->ansible)
  Downloading MarkupSafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Collecting attrs>=2.2.0 (from jsonschema==4.10.0->ansible-lint)
  Downloading attrs-25.4.0-py3-none-any.whl.metadata (10 kB)
Collecting jsonschema-specifications>=2023.03.6 (from jsonschema==4.10.0->ansible-lint)
  Downloading jsonschema-specifications-2025.9.1-py3-none-any.whl.metadata (2.9 kB)
Collecting rpds-py>=0.7.1 (from jsonschema==4.10.0->ansible-lint)
  Downloading rpds_py-0.28.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)
Collecting typing-extensions>=4.4.0 (from referencing==0.36.2->ansible-lint)
  Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting brace>=2.1.1 (from wcmatch==8.5.0->ansible-lint)
  Downloading brace-2.6-py3-none-any.whl.metadata (3.6 kB)
Downloading ansible-12.1.0-py3-none-any.whl (52.8 MB)
22.4/52.8 MB 89.5 MB/s 0:00:00
Downloading ansible-core-2.19.3-py3-none-any.whl (2.4 MB)
2.4/2.4 MB 60.6 MB/s 0:00:00
Downloading resolvelib-1.2.1-py3-none-any.whl (18 kB)
Downloading ansible-lint-25.9.2-py3-none-any.whl (322 kB)
Downloading yamllint-1.37.1-py3-none-any.whl (68 kB)
Downloading ansible-compat-25.8.2-py3-none-any.whl (27 kB)
Downloading black-25.9.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.manylinux_2_28_x86_64.whl (1.7 MB)
1.7/1.7 MB 75.5 MB/s 0:00:00
Downloading cffi-1.17.1-py3-none-any.whl (107 kB)
Downloading cryptography-46.0.3-cp311-abi3-manylinux_2_34_x86_64.whl (4.5 MB)
4.5/4.5 MB 90.1 MB/s 0:00:00
Downloading distro-1.9.0-py3-none-any.whl (20 kB)
Downloading filelock-3.20.0-py3-none-any.whl (16 kB)
Downloading importlib-metadata-8.7.0-py3-none-any.whl (27 kB)
Downloading Jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloading jsonschema-4.25.1-py3-none-any.whl (90 kB)
Downloading attrs-25.4.0-py3-none-any.whl (67 kB)
Downloading jsonschema-specifications-2025.9.1-py3-none-any.whl (18 kB)
Downloading MarkupSafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
Downloading mypy_extensions-1.1.0-py3-none-any.whl (5.0 kB)
Downloading packaging-25.0-py3-none-any.whl (66 kB)
Downloading pathspec-0.12.1-py3-none-any.whl (31 kB)
Downloading platformdirs-4.5.0-py3-none-any.whl (18 kB)
Downloading pytokens-0.2.0-py3-none-any.whl (12 kB)
Downloading PyYAML-6.0.3-cp312-cp312-manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (807 kB)
807.5/807.5 kB 99.3 MB/s 0:00:00
Downloading referencing-0.37.0-py3-none-any.whl (26 kB)
Downloading rpds_py-0.28.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (386 kB)
Downloading ruamel-yaml-0.18.16-py3-none-any.whl (119 kB)
Downloading ruamel-yaml-clib-0.2.14-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (753 kB)
753.1/753.1 kB 16.5 MB/s 0:00:00
Downloading subprocess-tee-0.4.2-py3-none-any.whl (5.2 kB)
Downloading wcmatch-10.1-py3-none-any.whl (39 kB)
Downloading typing-extensions-4.15.0-py3-none-any.whl (11 kB)
Downloading zipp-3.23.0-py3-none-any.whl (10 kB)
Downloading pycparser-2.23-py3-none-any.whl (118 kB)
WARNING: Ignoring invalid distribution -ip (/home/ubuntu/ansible-onboarding/.venv/lib/python3.12/site-packages)
Installing collected packages: zipp, typing-extensions, subprocess-tee, ruamel-yaml-clib, rpds-py, resolvelib, PyYAML, pytokens, pycparser, platformdirs, pathspec, packaging, mypy-extensions, MarkupSafe, filelock, distro, click, brace, attrs, yamllint, wcmatch, ruamel-yaml, referencing, Jinja2, importlib-metadata, cffi, black, jsonschema-specifications, cryptography, jsonschema, ansible-core, ansible-compat, ansible, ansible-lint
Successfully installed MarkupSafe-3.0.3 PyYAML-6.0.3 ansible-12.1.0 ansible-compat-25.8.2 ansible-core-2.19.3 ansible-lint-25.9.2 attrs-25.4.0 black-25.9.0 brace-2.6 cffi-1.17.1 cryptography-46.0.3 distro-1.9.0 filelock-3.20.0 importlib-metadata-8.7.0 Jinja2-3.1.6 jsonschema-4.25.1 jsonschema-specifications-2025.9.1 mypy-extensions-1.1.0 packaging-25.0 pathspec-0.12.1 platformdirs-4.5.0 pycparser-2.23 pytokens-0.2.0 referencing-0.37.0 resolvelib-1.2.1 rpds-py-0.28.0 ruamel-yaml-0.18.16 ruamel-yaml-clib-0.2.14 subprocess-tee-0.4.2 typing-extensions-4.15.0 wcmatch-10.1 yamllint-1.37.1 zipp-3.23.0
(venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

ALTERNATIVE CODE to install Ansible

```
python3 -m venv .venv
```

```
source .venv/bin/activate
```

```
pip install --upgrade pip
```

```
pip install ansible ansible-lint yamllint
```

Adriana Colaco

```
azureuser@ansiblevm:~/ansible-onboarding$ python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install ansible ansible-lint yamllint
Error: [Errno 13] Permission denied: '/home/azureuser/ansible-onboarding/.venv/bin/activate.fish'
Requirement already satisfied: pip in ./venv/lib/python3.12/site-packages (24.0)
Collecting pip
  Downloading pip-25.2-py3-none-any.whl.metadata (4.7 kB)
  Downloading pip-25.2-py3-none-any.whl (1.8 MB)
    1.8/1.8 MB 24.8 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.0
    Uninstalling pip-24.0:
      Successfully uninstalled pip-24.0
  WARNING: Failed to remove contents in a temporary directory '/home/azureuser/ansible-onboarding/.venv/lib/python3.12/site-packages/_ip-24.0.dist-info'.
  You can safely remove it manually.
  WARNING: Failed to remove contents in a temporary directory '/home/azureuser/ansible-onboarding/.venv/lib/python3.12/site-packages/_ip-'.
  You can safely remove it manually.
Successfully installed pip-25.2
WARNING: Ignoring invalid distribution -ip (/home/azureuser/ansible-onboarding/.venv/lib/python3.12/site-packages)
WARNING: Ignoring invalid distribution -ip (/home/azureuser/ansible-onboarding/.venv/lib/python3.12/site-packages)
Collecting ansible
  Downloading ansible-12.1.0-py3-none-any.whl.metadata (8.0 kB)
Collecting ansible-lint
  Downloading ansible_lint-25.9.2-py3-none-any.whl.metadata (6.3 kB)
Collecting yamllint
  Downloading yamllint-1.37.1-py3-none-any.whl.metadata (4.3 kB)
Collecting ansible-core==2.19.3 (from ansible)
  Downloading ansible_core-2.19.3-py3-none-any.whl.metadata (7.7 kB)
Collecting jinja2>=3.1.0 (from ansible-core==2.19.3->ansible)
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting PyYAML>=5.1 (from ansible-core==2.19.3->ansible)
  Downloading pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.4 kB)
Collecting cryptography (from ansible-core==2.19.3->ansible)
  Downloading cryptography-46.0.2-cp311-ab3-manlinux_2_34_x86_64.whl.metadata (5.7 kB)
Collecting packaging (from ansible-core==2.19.3->ansible)
  Downloading packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting resolvelib>=2.0.0,<=5.3 (from ansible-core==2.19.3->ansible)
  Downloading resolvelib-1.2.1-py3-none-any.whl.metadata (3.7 kB)
Collecting ansible-compat==25.8.2 (from ansible-lint)
  Downloading ansible_compat-25.8.2-py3-none-any.whl.metadata (3.4 kB)
Collecting black==24.3.0 (from ansible-lint)
  Downloading black-25.9.0-cp312-cp312-manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (83 kB)
Collecting cffi==1.17.1 (from ansible-lint)
  Downloading cffi-1.17.1-py3-none-any.whl.metadata (2.6 kB)
Collecting distro<1.9.0 (from ansible-lint)
  Downloading distro-1.9.0-py3-none-any.whl.metadata (6.8 kB)
Collecting filelock==3.8.2 (from ansible-lint)
  Downloading filelock-3.20.0-py3-none-any.whl.metadata (2.1 kB)
Collecting importlib-metadata==8.7.0 (from ansible-lint)
  Downloading importlib_metadata-8.7.0-py3-none-any.whl.metadata (4.8 kB)
Collecting jsonschema==4.10.0 (from ansible-lint)
  Downloading jsonschema-4.25.1-py3-none-any.whl.metadata (7.6 kB)
Collecting pathspec==10.3 (from ansible-lint)
  Downloading pathspec-0.12.1-py3-none-any.whl.metadata (21 kB)
Collecting referencing==0.36.2 (from ansible-lint)
  Downloading referencing-0.37.0-py3-none-any.whl.metadata (2.8 kB)
Collecting ruamel-yaml==0.18.11 (from ansible-lint)
  Downloading ruamel.yaml-0.18.15-py3-none-any.whl.metadata (25 kB)
Collecting ruamel-yaml-clib==0.2.12 (from ansible-lint)
  Downloading ruamel_yaml_clib-0.2.14-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.0 kB)
Collecting subprocess-tee==0.4.1 (from ansible-lint)
  Downloading subprocess_tee-0.4.2-py3-none-any.whl.metadata (3.3 kB)
Collecting wcmatch==8.5.0 (from ansible-lint)
  Downloading wcmatch-10.1-py3-none-any.whl.metadata (5.1 kB)
Collecting click==8.0.0 (from black==24.3.0->ansible-lint)
  Downloading click-8.3.0-py3-none-any.whl.metadata (2.6 kB)
Collecting mypy_extensions==0.4.3 (from black==24.3.0->ansible-lint)
  Downloading mypy_extensions-1.1.0-py3-none-any.whl.metadata (1.1 kB)
Collecting platformdirs==2 (from black==24.3.0->ansible-lint)
  Downloading platformdirs-4.5.0-py3-none-any.whl.metadata (12 kB)
Collecting pytokens==0.1.10 (from black==24.3.0->ansible-lint)
  Downloading pytokens-0.1.10-py3-none-any.whl.metadata (2.0 kB)
Collecting pycparser (from cffi==1.17.1->ansible-lint)
  Downloading pycparser-2.23-py3-none-any.whl.metadata (993 bytes)
Collecting zipp==3.20 (from importlib-metadata==8.7.0->ansible-lint)
  Downloading zipp-3.23.0-py3-none-any.whl.metadata (3.6 kB)
Collecting MarkupSafe==2.0 (from jinja2==3.1.0->ansible-core==2.19.3->ansible)
  Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Collecting attrs==22.2.0 (from jsonschema==4.10.0->ansible-lint)
  Downloading attrs-25.4.0-py3-none-any.whl.metadata (10 kB)
Collecting jsonschema-specifications==2023.03.6 (from jsonschema==4.10.0->ansible-lint)
  Downloading jsonschema_specifications-2025.9.1-py3-none-any.whl.metadata (2.9 kB)
Collecting rpds-py==0.7.1 (from jsonschema==4.10.0->ansible-lint)
  Downloading rpds_py-0.27.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.2 kB)
Collecting typing_extensions==4.4.0 (from referencing==0.36.2->ansible-lint)
  Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting brace2==2.1.1 (from wcmatch==8.5.0->ansible-lint)
  Downloading brace2-2.6-py3-none-any.whl.metadata (3.6 kB)
  Downloading ansible-12.1.0-py3-none-any.whl (52.8 MB)
    52.8/52.8 MB 18.2 MB/s 0:00:02
  Downloading ansible_core-2.19.3-py3-none-any.whl (2.4 MB)
    2.4/2.4 MB 98.3 MB/s 0:00:00
  Downloading resolvelib-1.2.1-py3-none-any.whl (18 kB)
  Downloading ansible_lint-25.9.2-py3-none-any.whl (322 kB)
  Downloading yamllint-1.37.1-py3-none-any.whl (68 kB)
  Downloading ansible_compat-25.8.2-py3-none-any.whl (27 kB)
  Downloading black-25.9.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.manylinux_2_28_x86_64.whl (1.7 MB)
    1.7/1.7 MB 83.1 MB/s 0:00:00
  Downloading cffi-1.17.1-py3-none-any.whl (219 kB)
  Downloading click-8.3.0-py3-none-any.whl (107 kB)
  Downloading cryptography-46.0.2-cp311-ab3-manlinux_2_34_x86_64.whl (4.5 MB)
    4.5/4.5 MB 101.5 MB/s 0:00:00
  Downloading distro-1.9.0-py3-none-any.whl (20 kB)
  Downloading filelock-3.20.0-py3-none-any.whl (16 kB)
  Downloading importlib_metadata-8.7.0-py3-none-any.whl (27 kB)
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
  Downloading jsonschema-4.25.1-py3-none-any.whl (90 kB)
  Downloading pathspec-10.3-py3-none-any.whl (67 kB)
  Downloading pytokens-0.1.10-py3-none-any.whl (18 kB)
  Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
  Downloading mypy_extensions-1.1.0-py3-none-any.whl (5.0 kB)
  Downloading packaging-25.0-py3-none-any.whl (66 kB)
  Downloading pathspec-0.12.1-py3-none-any.whl (31 kB)
  Downloading platformdirs-4.5.0-py3-none-any.whl (18 kB)
  Downloading pytokens-0.1.10-py3-none-any.whl (12 kB)
  Downloading pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (807 kB)
    807.0/807.0 kB 41.9 MB/s 0:00:00
  Downloading referencing-0.37.0-py3-none-any.whl (26 kB)
  Downloading rpds_py-0.27.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (386 kB)
  Downloading ruamel_yaml-0.18.15-py3-none-any.whl (119 kB)
  Downloading ruamel_yaml_clib-0.2.14-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (753 kB)
    753.1/753.1 kB 39.3 MB/s 0:00:00
  Downloading subprocess_tee-0.4.2-py3-none-any.whl (5.2 kB)
  Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
  Downloading wcmatch-10.1-py3-none-any.whl (59 kB)
  Downloading brace2-2.6-py3-none-any.whl (11 kB)
  Downloading zipp-3.23.0-py3-none-any.whl (10 kB)
  Downloading pycparser-2.23-py3-none-any.whl (118 kB)
  WARNING: Ignoring invalid distribution -ip (/home/azureuser/ansible-onboarding/.venv/lib/python3.12/site-packages)
Installing collected packages: zipp, typing-extensions, subprocess-tee, ruamel-yaml-clib, rpds-py, resolvelib, PyYAML, pytokens, pycparser, platformdirs, pathspec, packaging, mypy_extensions, MarkupSafe, filelock, distro, click, brace2, attrs, yamllint, wcmatch, ruamel-yaml, referencing, jinja2, importlib-metadata, cffi, black, jsonschema-specifications, cryptography, jsonschema, ema, ansible-core, ansible-compat, ansible, ansible-lint
  WARNING: Ignoring invalid distribution -ip (/home/azureuser/ansible-onboarding/.venv/lib/python3.12/site-packages)
Successfully installed MarkupSafe-3.0.3 PyYAML-6.0.3 ansible-12.1.0 ansible-compat-25.8.2 ansible-core-2.19.3 ansible-lint-25.9.2 attrs-25.4.0 brace2-2.6 brace2-2.6 cffi-1.17.1 click-8.3.0 cryptography-46.0.2 distro-1.9.0 filelock-3.20.0 importlib-metadata-8.7.0 jinja2-3.1.6 jsonschema-4.25.1 jsonschema-specifications-2025.9.1 mypy_extensions-1.1.0 packaging-25.0 pathspec-0.12.1 platformdirs-4.5.0 pycparser-2.23 pytokens-0.1.10 referencing-0.37.0 resolvelib-1.2.1 rpds-py-0.27.1 ruamel-yaml-0.18.15 ruamel-yaml-clib-0.2.14 subprocess-tee-0.4.2 typing-extensions-4.15.0 wcmatch-10.1 yamllint-1.37.1 zipp-3.23.0
(.venv) azureuser@ansiblevm:~/ansible-onboarding$
```

Verify installation

ansible --version

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ ansible --version
ansible [core 2.19.3]
  config file = None
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /home/ubuntu/ansible-onboarding/.venv/lib/python3.12/site-packages/ansible
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
  executable location = /home/ubuntu/ansible-onboarding/.venv/bin/ansible
  python version = 3.12.3 (main, Aug 14 2025, 17:47:21) [GCC 13.3.0] (/home/ubuntu/ansible-onboarding/.venv/bin/python)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

ansible-lint --version

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ ansible-lint --version
ansible-lint 25.9.2 using ansible-core:2.19.3 ansible-compat:25.8.2 ruamel-yaml:0.18.16 ruamel-yaml-clib:0.2.14
WARNING: Project directory /.ansible cannot be used for caching as it is not writable.
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Save dependencies

python -m pip freeze > requirements.txt

Alternative commands run

deactivate

cd .venv/lib/python3.12/site-packages

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ deactivate
ubuntu@ip-172-31-23-116:~/ansible-onboarding$ cd .venv/lib/python3.12/site-packages
ubuntu@ip-172-31-23-116:~/ansible-onboarding/.venv/lib/python3.12/site-packages$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

ls -la


```
ubuntu@ip-172-31-23-116:~/ansible-onboarding/.venv/lib/python3.12/site-packages$ ls -la
total 7384
drwxrwxr-x 77 ubuntu ubuntu 4096 Oct 27 13:48 .
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:26 ..
-rwxrwxr-x 1 ubuntu ubuntu 4113144 Oct 27 13:47 30fcd23745efef32ce681_mypyc.cpython-312-x86_64-linux-gnu.so
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 27 13:47 _pycache_
-rw-rw-r-- 1 ubuntu ubuntu 19 Oct 27 13:47 _black_version.py
-rw-rw-r-- 1 ubuntu ubuntu 13 Oct 27 13:47 _black_version.pyi
-rwxrwxr-x 1 ubuntu ubuntu 348808 Oct 27 13:47 _cffi_backend.cpython-312-x86_64-linux-gnu.so
-rw-rw-r-- 1 ubuntu ubuntu 2592192 Oct 27 13:47 ruamel.yaml.cpython-312-x86_64-linux-gnu.so
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 _yaml_
drwxrwxr-x 21 ubuntu ubuntu 4096 Oct 27 13:47 ansible
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:48 ansible-12.1.0.dist-info
drwxrwxr-x 48 ubuntu ubuntu 4096 Oct 27 13:47 ansible_collections
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 ansible_compat
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 ansible_compat-25.8.2.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 ansible_core-2.19.3.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:48 ansible_lint-25.9.2.dist-info
drwxrwxr-x 7 ubuntu ubuntu 4096 Oct 27 13:47 ansible_test
drwxrwxr-x 9 ubuntu ubuntu 4096 Oct 27 13:48 ansiblelint
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 attr
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 attrrs
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 attrrs-25.4.0.dist-info
drwxrwxr-x 4 ubuntu ubuntu 4096 Oct 27 13:47 black
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 black-25.9.0.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 blackd
drwxrwxr-x 4 ubuntu ubuntu 4096 Oct 27 13:47 blib2to3
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 brace
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 brace-2.6.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 cffi
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 cffi-2.0.0.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 click
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 click-8.3.0.dist-info
drwxrwxr-x 5 ubuntu ubuntu 4096 Oct 27 13:47 cryptography
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 cryptography-46.0.3.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 distro
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 27 13:47 distro-1.9.0.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 filelock
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 filelock-3.20.0.dist-info
drwxrwxr-x 4 ubuntu ubuntu 4096 Oct 27 13:47 importlib_metadata
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 importlib_metadata-8.7.0.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 Jinja2
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 Jinja2-3.1.6.dist-info
drwxrwxr-x 5 ubuntu ubuntu 4096 Oct 27 13:47 jsonschema
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 jsonschema-4.25.1.dist-info
drwxrwxr-x 5 ubuntu ubuntu 4096 Oct 27 13:47 jsonschema_specifications
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 jsonschema_specifications-2025.9.1.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 markupsafe
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 markupsafe-3.0.3.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 mypy_extensions-1.1.0.dist-info
-rw-rw-r-- 1 ubuntu ubuntu 7754 Oct 27 13:47 mypy_extensions.py
drwxrwxr-x 4 ubuntu ubuntu 4096 Oct 27 13:47 packaging
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 27 13:47 packaging-25.0.dist-info
drwxrwxr-x 4 ubuntu ubuntu 4096 Oct 27 13:47 pathspec
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 27 13:47 pathspec-0.12.1.dist-info
drwxrwxr-x 5 ubuntu ubuntu 4096 Oct 27 13:40 pip
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:40 pip-25.3.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 platformdirs
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 platformdirs-4.5.0.dist-info
drwxrwxr-x 4 ubuntu ubuntu 4096 Oct 27 13:47 pycparser
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 27 13:47 pycparser-2.23.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 pytokens
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 27 13:47 pytokens-0.2.0.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 pyyaml-6.0.3.dist-info
drwxrwxr-x 4 ubuntu ubuntu 4096 Oct 27 13:47 referencing
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 referencing-0.37.0.dist-info
drwxrwxr-x 4 ubuntu ubuntu 4096 Oct 27 13:47 resolvelib
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 resolvelib-1.2.1.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 rpds
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 rpds_py-0.28.0.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 ruamel
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 ruamel.yaml-0.18.16.dist-info
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 27 13:47 ruamel.yaml.clib-0.2.14.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 subprocess_tee
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 27 13:47 subprocess_tee-0.4.2.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 typing_extensions-4.15.0.dist-info
-rw-rw-r-- 1 ubuntu ubuntu 160429 Oct 27 13:47 typing_extensions.py
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 wcmatch
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 27 13:47 wcmatch-10.1.dist-info
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 yaml
drwxrwxr-x 5 ubuntu ubuntu 4096 Oct 27 13:47 yamllint
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 yamllint-1.37.1.dist-info
drwxrwxr-x 4 ubuntu ubuntu 4096 Oct 27 13:47 zipp
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 27 13:47 zipp-3.23.0.dist-info
drwxr-xr-x 5 root root 4096 Oct 27 13:31 ~ip
drwxr-xr-x 2 root root 4096 Oct 27 13:31 ~ip-24.0.dist-info'
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

ls -d ~*

```
ubuntu@ip-172-31-23-116:~/ansible-onboarding/.venv/lib/python3.12/site-packages$ ls -d ~*
'~ip' '~ip-24.0.dist-info'
```

sudo rm -rf ~ip*

```
ubuntu@ip-172-31-23-116:~/ansible-onboarding/.venv/lib/python3.12/site-packages$ sudo rm -rf ~ip*
ubuntu@ip-172-31-23-116:~/ansible-onboarding/.venv/lib/python3.12/site-packages$
```

cd ..

cd ..

cd ..

cd ..

source .venv/bin/activate

```
ubuntu@ip-172-31-23-116:~/ansible-onboarding/.venv/lib/python3.12/site-packages$ cd ..
ubuntu@ip-172-31-23-116:~/ansible-onboarding/.venv/lib/python3.12$ cd ..
ubuntu@ip-172-31-23-116:~/ansible-onboarding/.venv/lib$ cd ..
ubuntu@ip-172-31-23-116:~/ansible-onboarding/.venv$ source .venv/bin/activate
-bash: .venv/bin/activate: No such file or directory
ubuntu@ip-172-31-23-116:~/ansible-onboarding/.venv$ ls
bin  include  lib  lib64  pyvenv.cfg
ubuntu@ip-172-31-23-116:~/ansible-onboarding/.venv$ cd ..
ubuntu@ip-172-31-23-116:~/ansible-onboarding$ source .venv/bin/activate
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

pip freeze > requirements.txt

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ pip freeze > requirements.txt
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ ls
requirements.txt
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ vim requirements.txt
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ vim requirements.txt
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116


```
ansible==12.1.0
ansible-compat==25.8.2
ansible-core==2.19.3
ansible-lint==25.9.2
attrs==25.4.0
black==25.9.0
bracex==2.6
cffi==2.0.0
click==8.3.0
cryptography==46.0.3
distro==1.9.0
filelock==3.20.0
importlib_metadata==8.7.0
Jinja2==3.1.6
jsonschema==4.25.1
jsonschema-specifications==2025.9.1
MarkupSafe==3.0.3
mypy_extensions==1.1.0
packaging==25.0
pathspec==0.12.1
platformdirs==4.5.0
pycparser==2.23
pytokens==0.2.0
PyYAML==6.0.3
referencing==0.37.0
resolvelib==1.2.1
rpds-py==0.28.0
ruamel.yaml==0.18.16
ruamel.yaml.clib==0.2.14
subprocess-tee==0.4.2
typing_extensions==4.15.0
"requirements.txt" 34L, 625B
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

✓ Result:

You have a Python virtual environment with Ansible and lint tools installed.

📁 2. VS Code Setup (Tooling)

Application to Use:

- **Visual Studio Code (VS Code)**
- Install these **extensions** from the Marketplace:
 -  Red Hat Ansible — [link](#)
 -  YAML (by Red Hat)
 -  Python (by Microsoft)
 -  EditorConfig (recommended)

Steps:

In the project folder (~/.ansible-onboarding), create:

.vscode/settings.json

```
{  
  "python.defaultInterpreterPath": "${workspaceFolder}/.venv/bin/python",  
  "ansible.python.interpreterPath": "${workspaceFolder}/.venv/bin/python",  
  "ansibleLint.enabled": true,  
  "yaml.validate": true,  
  "files.trimTrailingWhitespace": true,  
  "editor.formatOnSave": true  
}
```

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ mkdir -p .vscode  
nano .vscode/settings.json  
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

```
GNU nano 7.2 .vscode/settings.json
{
  "python.defaultInterpreterPath": "${workspaceFolder}/.venv/bin/python",
  "ansible.python.interpreterPath": "${workspaceFolder}/.venv/bin/python",
  "ansibleLint.enabled": true,
  "yaml.validate": true,
  "files.trimTrailingWhitespace": true,
  "editor.formatOnSave": true
}
```

.editorconfig

root = true

[*]

charset = utf-8

end_of_line = lf

indent_style = space

indent_size = 2

insert_final_newline = true

trim_trailing_whitespace = true

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ nano .editorconfig
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

```
GNU nano 7.2 .editorconfig
root = true

[*]
charset = utf-8
end_of_line = lf
indent_style = space
indent_size = 2
insert_final_newline = true
trim_trailing_whitespace = true
```



Result:

VS Code automatically uses your Python venv, lints YAML files, and formats code on save.

3. Create Baseline ansible.cfg

Application to Use:

- VS Code or Terminal text editor (nano/vim)

Create ansible.cfg in project root:

[defaults]

inventory = inventories/

roles_path = roles:./ansible/roles

host_key_checking = True

retry_files_enabled = False

interpreter_python = auto_silent

forks = 10

timeout = 30

stdout_callback = yaml

bin_ansible_callbacks = True

[ssh_connection]

pipelining = True

ssh_args = -o ControlMaster=auto -o ControlPersist=60s

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ nano ansible.cfg  
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

```
GNU nano 7.2 ansible.cfg  
[defaults]  
inventory = inventories/  
roles_path = roles:./ansible/roles  
host_key_checking = True  
retry_files_enabled = False  
interpreter_python = auto_silent  
forks = 10  
timeout = 30  
stdout_callback = yaml  
bin_ansible_callbacks = True  
  
[ssh_connection]  
pipelining = True  
ssh_args = -o ControlMaster=auto -o ControlPersist=60s
```

✓ Result:

You have a standard Ansible configuration file aligned with team standards.

🔒 4. SSH Readiness

Application to Use:

- Terminal

Steps:

Generate SSH key

ssh-keygen -t ed25519 -C "you@company" -N "" -f ~/.ssh/id_ed25519

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ ssh-keygen -t ed25519 -C "adrina@company.com" -N "" -f ~/.ssh/id_ed25519
Generating public/private ed25519 key pair.
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256: [REDACTED] /JIVwchD+o adrina@company.com
The key's randomart image is:
+--[ED25519 256]--+
|  o .               |
| o = o             |
| . = + .           |
| E + = S           |
| [REDACTED]         |
|o B = o ++o       |
| o O . = O*o.     |
| [REDACTED]         |
+----[SHA256]-----+
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Start SSH agent and add your key

eval "\$(ssh-agent -s)"

ssh-add ~/.ssh/id_ed25519

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519
Agent pid 5054
Identity added: /home/ubuntu/.ssh/id_ed25519 (adrina@company.com)
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Create SSH config file

nano ~/.ssh/config

Paste this inside:

Host *

ServerAliveInterval 30

ServerAliveCountMax 4

AddKeysToAgent yes

IdentityFile ~/.ssh/id_ed25519

StrictHostKeyChecking ask

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ nano ~/.ssh/config  
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

```
GNU nano 7.2 /home/ubuntu/.ssh/config  
  
Host *  
  ServerAliveInterval 30  
  ServerAliveCountMax 4  
  AddKeysToAgent yes  
  IdentityFile ~/.ssh/id_ed25519  
  StrictHostKeyChecking ask  
  
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location j M-U Undo
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116



Result:

You have a working SSH key and agent ready for secure connections.



5. Git Identity, Signing & Hooks

Application to Use:

- **Git (CLI)**
- **GitHub / GitLab (optional for signing)**

Steps:

git config --global user.name "Your Name"

git config --global user.email "your.email@company.com"

git config --global init.defaultBranch main

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ git config --global user.name "Adrina"  
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ git config --global user.email "adrina@company.com"  
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ git config --global init.defaultBranch main
```

(Optional) Enable GPG or SSH commit signing (follow your company's documentation).

SSH commit signing

git config --global gpg.format ssh

git config --global user.signingkey ~/.ssh/id_ed25519.pub

git config --global commit.gpgsign true

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ git config --global gpg.format ssh  
git config --global user.signingkey ~/.ssh/id_ed25519.pub  
git config --global commit.gpgsign true  
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Install Pre-Commit Hooks:

❓ **Deactivate any active environment** (if one is active):

Deactivate

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ deactivate  
ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

❓ **Remove your existing venv** (if it is broken or misconfigured):


```
rm -rf .venv
```

```
ubuntu@ip-172-31-23-116:~/ansible-onboarding$ rm -rf .venv
ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

❓ Install python3-venv (if not already installed):

On Ubuntu / Debian:

```
sudo apt update
```

```
sudo apt install python3-venv
```

```
aws
[Alt+S]
United States (N. Virginia)
Account ID: 3111-4154-2113
Adrina Colaco

ubuntu@ip-172-31-23-116:~/ansible-onboarding$ sudo apt update
sudo apt install python3-venv
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [377 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7168 B]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:16 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Fetched 1024 kB in 1s (1055 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
45 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-venv is already the newest version (3.12.3-0ubuntu2).
0 upgraded, 0 newly installed, 0 to remove and 45 not upgraded.
ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

CloudShell Feedback

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❓ Create a fresh virtual environment:

```
python3 -m venv .venv
```

❓ Activate it:

```
source .venv/bin/activate
```

```
ubuntu@ip-172-31-23-116:~/ansible-onboarding$ python3 -m venv .venv
ubuntu@ip-172-31-23-116:~/ansible-onboarding$ source .venv/bin/activate
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

🔗 Check that pip is pointing inside the venv:

which pip

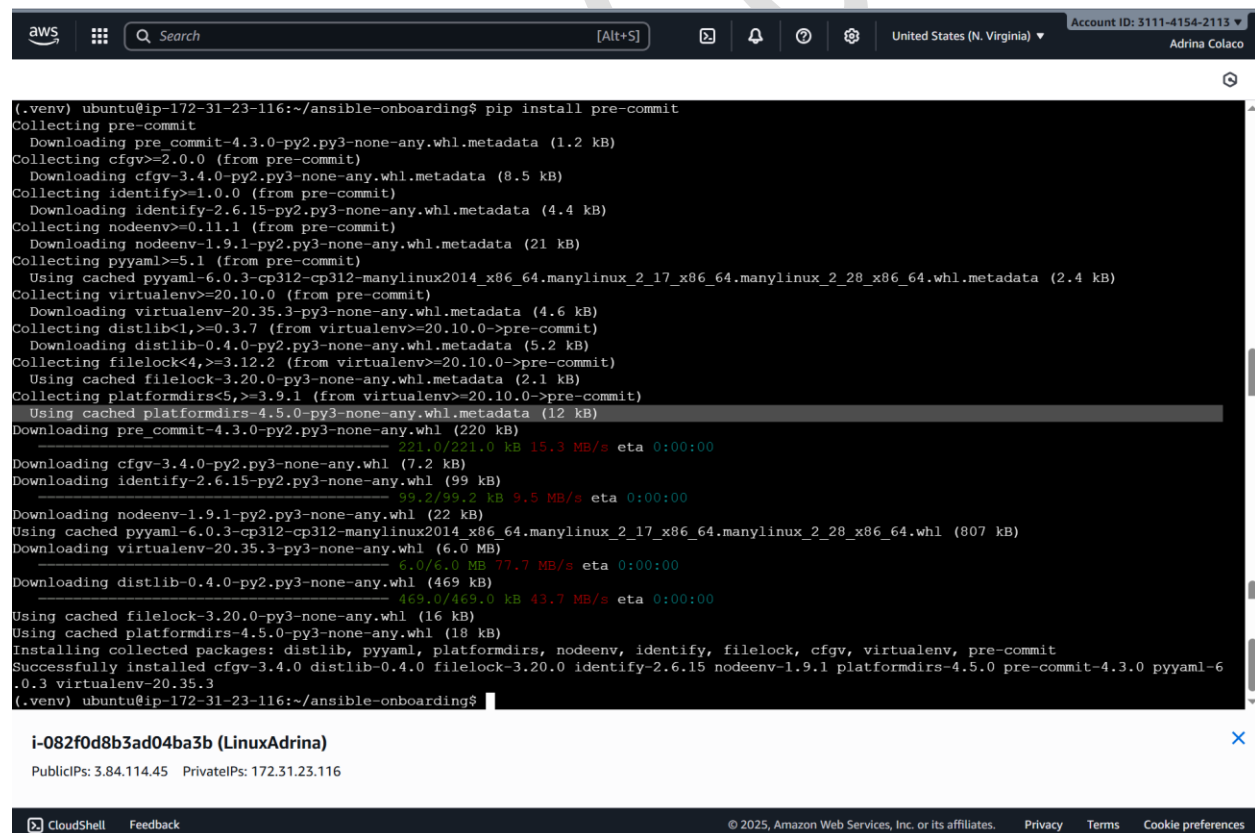
should show something like /home/azureuser/ansible-onboarding/.venv/bin/pip

pip --version

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ which pip
/home/ubuntu/ansible-onboarding/.venv/bin/pip
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ pip --version
pip 24.0 from /home/ubuntu/ansible-onboarding/.venv/lib/python3.12/site-packages/pip (python 3.12)
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

🔗 Install pre-commit (and other tools) (without sudo):

pip install pre-commit



```
aws [Search] [Alt+S] United States (N. Virginia) Account ID: 3111-4154-2113 Adrina Colaco

(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ pip install pre-commit
Collecting pre-commit
  Downloading pre_commit-4.3.0-py2.py3-none-any.whl.metadata (1.2 kB)
Collecting cfgv<=2.0.0 (from pre-commit)
  Downloading cfgv-3.4.0-py2.py3-none-any.whl.metadata (8.5 kB)
Collecting identify<=1.0.0 (from pre-commit)
  Downloading identify-2.6.15-py2.py3-none-any.whl.metadata (4.4 kB)
Collecting nodeenv<=0.11.1 (from pre-commit)
  Downloading nodeenv-1.9.1-py2.py3-none-any.whl.metadata (21 kB)
Collecting pyyaml<=5.1 (from pre-commit)
  Using cached pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.4 kB)
Collecting virtualenv<=20.10.0 (from pre-commit)
  Downloading virtualenv-20.35.3-py3-none-any.whl.metadata (4.6 kB)
Collecting distlib<1,>=0.3.7 (from virtualenv<=20.10.0->pre-commit)
  Downloading distlib-0.4.0-py2.py3-none-any.whl.metadata (5.2 kB)
Collecting filelock<4,>=3.12.2 (from virtualenv<=20.10.0->pre-commit)
  Using cached filelock-3.20.0-py3-none-any.whl.metadata (2.1 kB)
Collecting platformdirs<5,>=3.9.1 (from virtualenv<=20.10.0->pre-commit)
  Using cached platformdirs-4.5.0-py3-none-any.whl.metadata (12 kB)
Downloading pre_commit-4.3.0-py2.py3-none-any.whl (220 kB)
221.0/221.0 kB 15.3 MB/s eta 0:00:00
Downloading cfgv-3.4.0-py2.py3-none-any.whl (7.2 kB)
Downloading identify-2.6.15-py2.py3-none-any.whl (99 kB)
99.2/99.2 kB 9.5 MB/s eta 0:00:00
Downloading nodeenv-1.9.1-py2.py3-none-any.whl (22 kB)
Using cached pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (807 kB)
Downloading virtualenv-20.35.3-py3-none-any.whl (6.0 MB)
6.0/6.0 MB 77.7 MB/s eta 0:00:00
Downloading distlib-0.4.0-py2.py3-none-any.whl (469 kB)
469.0/469.0 kB 43.7 MB/s eta 0:00:00
Using cached filelock-3.20.0-py3-none-any.whl (16 kB)
Using cached platformdirs-4.5.0-py3-none-any.whl (18 kB)
Installing collected packages: distlib, pyyaml, platformdirs, nodeenv, identify, filelock, cfgv, virtualenv, pre-commit
Successfully installed cfgv-3.4.0 distlib-0.4.0 filelock-3.20.0 identify-2.6.15 nodeenv-1.9.1 platformdirs-4.5.0 pre-commit-4.3.0 pyyaml-6.0.3 virtualenv-20.35.3
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

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Create .pre-commit-config.yaml:

repos:

- repo: <https://github.com/adrienverge/yamllint>

rev: v1.35.1

hooks:

- id: yamllint

- repo: <https://github.com/ansible/ansible-lint>

rev: v24.6.1

hooks:

- id: ansible-lint

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ touch .pre-commit-config.yaml
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ vim .pre-commit-config.yaml
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Activate hooks:

pre-commit install

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ git init
Initialized empty Git repository in /home/ubuntu/ansible-onboarding/.git/
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ git --version
git version 2.43.0
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ pre-commit install
pre-commit installed at .git/hooks/pre-commit
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116



Result:

Every commit automatically runs YAML and Ansible linting for quality checks.



6. README + Checklist

Application to Use:

- VS Code

Create README.md describing:

Sections to Include:

- OS & Python version
- How Ansible was installed
- Installed VS Code extensions
- SSH key setup
- Git configuration
- How to activate venv
- How to run ansible-lint / yamllint
- Notes for corporate network or proxy
- “New Machine? Do This” checklist (10–12 quick setup steps)

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ nano README.md
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

```
GNU nano 7.2                                README.md
Ansible Onboarding Project

This project sets up and configures an Ansible development environment on an Ubuntu AWS VM, following best practices for Python environments, linting, and Git commit signing.

System Information
Component      Version / Details
OS             Ubuntu 22.04 LTS (AWS EC2 VM)
Python        3.10+
Ansible        Installed via pip inside virtual environment
Text Editor / IDE  VS Code (via Remote SSH)
Installation Steps
1 Create project folder and virtual environment
cd ~
mkdir -p ansible-onboarding
cd ansible-onboarding

sudo apt update
sudo apt install python3-venv -y

python3 -m venv .venv
source .venv/bin/activate

2 Install required packages
pip install ansible ansible-lint yamllint

3 Create configuration files
ansible.cfg - project-level config

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute   ^C Location  ^U Undo      ^M Set Mark  ^J To Bracket ^P Previous  ^B Back
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify   ^_ Go To Line ^- Redo      ^C Copy      ^_ Where Was  ^N Next     ^S For
```

i-082f0d8b3ad04ba3b (LinuxAdrina)
PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

✓ 7. Final Verification

Run the following to confirm setup:

ansible --version

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ ansible --version
ansible [core 2.19.3]
  config file = /home/ubuntu/ansible-onboarding/ansible.cfg
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /home/ubuntu/ansible-onboarding/.venv/lib/python3.12/site-packages/ansible
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
  executable location = /home/ubuntu/ansible-onboarding/.venv/bin/ansible
  python version = 3.12.3 (main, Aug 14 2025, 17:47:21) [GCC 13.3.0] (/home/ubuntu/ansible-onboarding/.venv/bin/python)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

ansible-lint --version

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ ansible-lint --version
ansible-lint 25.9.2 using ansible-core:2.19.3 ansible-compat:25.8.2 ruamel-yaml:0.18.16 ruamel-yaml-clib:0.2.14
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

pre-commit run --all-files

ssh-add -l

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ pre-commit run --all-files
[INFO] Initializing environment for https://github.com/adrienverge/yamllint.
[INFO] Initializing environment for https://github.com/ansible/ansible-lint.
[INFO] Initializing environment for https://github.com/ansible/ansible-lint:ansible-core>=2.13.3.
[INFO] Installing environment for https://github.com/adrienverge/yamllint.
[INFO] Once installed this environment will be reused.
[INFO] This may take a few minutes...
[INFO] Installing environment for https://github.com/ansible/ansible-lint.
[INFO] Once installed this environment will be reused.
[INFO] This may take a few minutes...
yamllint.....(no files to check)Skipped
Ansible-lint.....Failed
- hook id: ansible-lint
- exit code: 1

Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "/home/ubuntu/.cache/pre-commit/repol8k6f8/py_env-python3.12/lib/python3.12/site-packages/ansiblelint/_main_.py", line 50, in <module>
    from ansiblelint import cli
  File "/home/ubuntu/.cache/pre-commit/repol8k6f8/py_env-python3.12/lib/python3.12/site-packages/ansiblelint/cli.py", line 30, in <module>
    from ansiblelint.yaml_utils import clean_json
  File "/home/ubuntu/.cache/pre-commit/repol8k6f8/py_env-python3.12/lib/python3.12/site-packages/ansiblelint/yaml_utils.py", line 34, in <module>
    from ansiblelint.utils import Task
  File "/home/ubuntu/.cache/pre-commit/repol8k6f8/py_env-python3.12/lib/python3.12/site-packages/ansiblelint/utils.py", line 45, in <module>
    from ansible.parsing.yaml.constructor import AnsibleConstructor, AnsibleMapping
ModuleNotFoundError: No module named 'ansible.parsing.yaml.constructor'
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

✓ Expected Output:

- All commands run without errors.
- ansible-onboarding/ folder contains:
- .venv/

- requirements.txt
- .vscode/settings.json
- .editorconfig
- ansible.cfg
- .pre-commit-config.yaml

```
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$ ls -la
total 44
drwxrwxr-x 6 ubuntu ubuntu 4096 Oct 27 17:28 .
drwxr-x--- 7 ubuntu ubuntu 4096 Oct 27 17:21 ..
drwxrwxr-x 5 ubuntu ubuntu 4096 Oct 27 17:29 .ansible
-rw-rw-r-- 1 ubuntu ubuntu 147 Oct 27 15:13 .editorconfig
drwxrwxr-x 7 ubuntu ubuntu 4096 Oct 27 17:13 .git
-rw-rw-r-- 1 ubuntu ubuntu 209 Oct 27 17:12 .pre-commit-config.yaml
drwxrwxr-x 6 ubuntu ubuntu 4096 Oct 27 17:28 .venv
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 27 15:11 .vscode
-rw-rw-r-- 1 ubuntu ubuntu 3625 Oct 27 17:17 README.md
-rw-rw-r-- 1 ubuntu ubuntu 387 Oct 27 15:24 ansible.cfg
-rw-rw-r-- 1 ubuntu ubuntu 625 Oct 27 14:21 requirements.txt
(.venv) ubuntu@ip-172-31-23-116:~/ansible-onboarding$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

README.md

```
GNU nano 7.2                                README.md
Ansible Onboarding Project

This project sets up and configures an Ansible development environment on an Ubuntu AWS VM, following best practices for Python environments, linting, and Git commit signing.

System Information
Component      Version / Details
OS             Ubuntu 22.04 LTS (AWS EC2 VM)
Python        3.10+
Ansible        Installed via pip inside virtual environment
Text Editor / IDE  VS Code (via Remote SSH)

Installation Steps
1 Create project folder and virtual environment
cd ~
mkdir -p ansible-onboarding
cd ansible-onboarding

sudo apt update
sudo apt install python3-venv -y

python3 -m venv .venv
source .venv/bin/activate

2 Install required packages
pip install ansible ansible-lint yamllint

3 Create configuration files
ansible.cfg - project-level config

[ Read 147 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^M Set Mark   ^_ To Bracket  ^P Previous   ^B Back
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^V Go To Line ^S Redo       ^- Copy       ^G Where Was  ^N Next       ^F For
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Assignment 34 – Ad-Hoc Automation on Azure- 4 VMs, Inventory & Passwordless SSH

Project Overview

You'll:

1. Use **Terraform** to create 4 Ubuntu EC2 instances on AWS.
2. Set up **passwordless SSH** to all instances.
3. Create an **Ansible inventory** with groups (web, app, db).
4. Run **ad-hoc Ansible commands** to test connectivity, install packages, and manage services.

Folder Structure

```
aws-fleet/  
├─ main.tf  
├─ variables.tf  
├─ outputs.tf  
├─ inventory.ini  
└─ setup.sh
```

```
ubuntu@ip-172-31-23-116:~$ mkdir aws-fleet  
ubuntu@ip-172-31-23-116:~$ cd aws-fleet/  
ubuntu@ip-172-31-23-116:~/aws-fleet$ nano main.tf  
ubuntu@ip-172-31-23-116:~/aws-fleet$ nano variables.tf  
ubuntu@ip-172-31-23-116:~/aws-fleet$ nano outputss.tf  
ubuntu@ip-172-31-23-116:~/aws-fleet$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Step 1: Terraform Configuration (4 EC2 VMs)

main.tf

```
provider "aws" {
  region = "us-east-1"
}

resource "aws_key_pair" "my_key" {
  key_name = "my-keypair"
  public_key = file("~/ssh/id_ed25519.pub")
}

resource "aws_security_group" "fleet_sg" {
  name      = "fleet-sg"
  description = "Allow SSH and HTTP"

  ingress {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port = 80
    to_port   = 80
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port = 0
    to_port   = 0
    protocol  = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}

variable "instance_count" {
```



```
default = 4
}

resource "aws_instance" "fleet" {
  count      = var.instance_count
  ami       = "ami-0fc5d935ebf8bc3bc" # Ubuntu 22.04 LTS (check your region)
  instance_type = "t2.micro"
  key_name   = aws_key_pair.my_key.key_name
  vpc_security_group_ids = [aws_security_group.fleet_sg.id]

  tags = {
    Name = "fleet-${count.index}"
  }

  provisioner "remote-exec" {
    inline = [
      "sudo apt update -y",
      "sudo apt install -y python3"
    ]
  }

  connection {
    type      = "ssh"
    user      = "ubuntu"
    private_key = file("~/ssh/id_ed25519")
    host      = self.public_ip
  }
}

output "public_ips" {
  value = aws_instance.fleet[*].public_ip
}
```

```
GNU nano 7.2 main.tf
provider "aws" {
  region = "us-east-1"
}

resource "aws_key_pair" "my_key" {
  key_name   = "my-keypair"
  public_key = file("~/ssh/id_ed25519.pub")
}

resource "aws_security_group" "fleet_sg" {
  name        = "fleet-sg"
  description = "Allow SSH and HTTP"

  ingress {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port = 80
    to_port   = 80
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {

i-082f0d8b3ad04ba3b (LinuxAdrina)
PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116
```

variables.tf

```
variable "aws_region" {
  default = "us-east-1"
}
```

```
GNU nano 7.2 variables.tf
variable "aws_region" {
  default = "us-east-1"
}
```

outputs.tf

```
output "vm_ips" {
  value = aws_instance.fleet[*].public_ip
  description = "Public IPs of all 4 EC2 instances"
```

```
}
```

```
GNU nano 7.2                                     outputss.tf
output "vm_ips" {
  value = aws_instance.fleet[*].public_ip
  description = "Public IPs of all 4 EC2 instances"
}
```

Step 2: Deploy with Terraform

terraform init

```
ubuntu@ip-172-31-23-116:~/aws-fleet$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.19.0...
- Installed hashicorp/aws v6.19.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
ubuntu@ip-172-31-23-116:~/aws-fleet$
```

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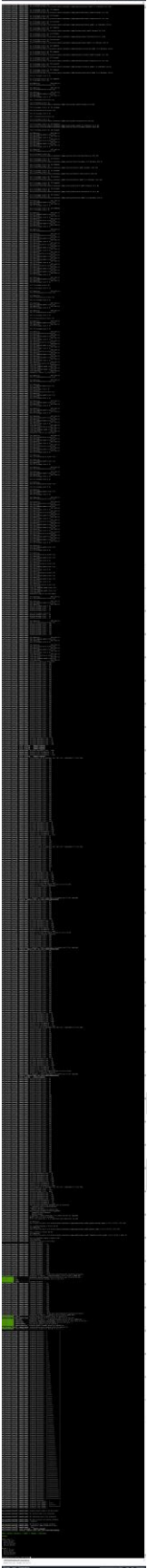
PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

terraform plan

[illegible]

terraform apply -auto-approve

Adriana Colaco



Adriana Colaco

Once done:

terraform output vm_ips

You'll get something like:

```
[  
  "3.92.12.10",  
  "54.81.15.32",  
  "18.208.4.56",  
  "44.202.97.22"  
]
```

Apply complete! Resources: 5 added, 0 changed, 0 destroyed.

Outputs:

```
public_ips = [  
  "50.17.60.116",  
  "52.91.23.50",  
  "35.172.183.242",  
  "54.204.128.25",  
]  
vm_ips = [  
  "50.17.60.116",  
  "52.91.23.50",  
  "35.172.183.242",  
  "54.204.128.25",  
]  
ubuntu@ip-172-31-23-116:~/aws-fleet$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Step 3: Passwordless SSH Setup

Run this shell script:

```
#!/bin/bash
```

```
# setup.sh
```

```
IPs=( "50.17.60.116" "52.91.23.50" "35.172.183.242" "54.204.128.25")
```

for ip in "\${IPs[@]}"; do

ssh-copy-id -i ~/.ssh/id_ed25519.pub ubuntu@\$ip

done

```
aws
[Alt+S]
United States (N. Virginia)
Account ID: 3111-4154-2113
Adrina Colaco

ubuntu@ip-172-31-23-116:~/aws-fleet$ #!/bin/bash
# setup.sh
IPs=( "50.17.60.116" "52.91.23.50" "35.172.183.242" "54.204.128.25" )

for ip in "${IPs[@]}"; do
    ssh-copy-id -i ~/.ssh/id_ed25519.pub ubuntu@$ip
done

/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ubuntu/.ssh/id_ed25519.pub"
The authenticity of host '50.17.60.116 (50.17.60.116)' can't be established.
ED25519 key fingerprint is SHA256:9hEaCmf3b5WlH/VqiAHD4cD9Xp+46iUstVNH3Q0Qg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
(if you think this is a mistake, you may want to use -f option)

/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ubuntu/.ssh/id_ed25519.pub"
The authenticity of host '52.91.23.50 (52.91.23.50)' can't be established.
ED25519 key fingerprint is SHA256:8SUG57Xruttredj7Bg0464QoslgAP5r0UboigHnabjQ8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
(if you think this is a mistake, you may want to use -f option)

/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ubuntu/.ssh/id_ed25519.pub"
The authenticity of host '35.172.183.242 (35.172.183.242)' can't be established.
ED25519 key fingerprint is SHA256:1zi3T2k61lBzKJAFg//xhls/RMzzQKLE3lQrGK2fyA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
(if you think this is a mistake, you may want to use -f option)

/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ubuntu/.ssh/id_ed25519.pub"
The authenticity of host '54.204.128.25 (54.204.128.25)' can't be established.
ED25519 key fingerprint is SHA256:AV8ymG0ll6/fdUHVqItxtFcIdmVNZAmM9lUem8bzeW9Dg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
(if you think this is a mistake, you may want to use -f option)
```

Then test:

ssh ubuntu@50.17.60.116

If no password is asked, it's working.


```
ubuntu@ip-172-31-23-116:~/aws-fleet$ ssh ubuntu@52.91.23.50
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 6.2.0-1012-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Fri Oct 31 09:49:02 UTC 2025

System load:  0.0               Processes:            98
Usage of /:   23.9% of 7.57GB   Users logged in:     0
Memory usage: 23%              IPv4 address for eth0: 172.31.19.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

176 updates can be applied immediately.
176 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Oct 31 09:04:30 2025 from 3.84.114.45
ubuntu@ip-172-31-19-4:~$ exit
logout
Connection to 52.91.23.50 closed.
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Step 4: Ansible Inventory File

inventory.ini

[web]

50.17.60.116

52.91.23.50

[app]

35.172.183.242

[db]

54.204.128.25

[all:vars]

ansible_user=ubuntu

ansible_ssh_private_key_file=~/.ssh/id_ed25519

```
ubuntu@ip-172-31-23-116:~/aws-fleet$ nano inventory.ini
ubuntu@ip-172-31-23-116:~/aws-fleet$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

```
GNU nano 7.2 inventory.ini
[web]
50.17.60.116
52.91.23.50

[app]
35.172.183.242

[db]
54.204.128.25

[all:vars]
ansible_user=ubuntu
ansible_ssh_private_key_file=~/.ssh/id_ed25519
```

Step 5: Run Ad-Hoc Ansible Commands

Test connectivity

ansible all -i inventory.ini -m ping

```
ubuntu@ip-172-31-23-116:~/aws-fleet$ ansible all -i inventory.ini -m ping
54.204.128.25 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
35.172.183.242 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
50.17.60.116 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
52.91.23.50 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu@ip-172-31-23-116:~/aws-fleet$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Check current user

```
ansible all -i inventory.ini -a "whoami"
```

```
ubuntu@ip-172-31-23-116:~/aws-fleet$ ansible all -i inventory.ini -a "whoami"
35.172.183.242 | CHANGED | rc=0 >>
ubuntu
52.91.23.50 | CHANGED | rc=0 >>
ubuntu
50.17.60.116 | CHANGED | rc=0 >>
ubuntu
54.204.128.25 | CHANGED | rc=0 >>
ubuntu
ubuntu@ip-172-31-23-116:~/aws-fleet$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Check uptime

ansible all -i inventory.ini -a "uptime"

```
ubuntu@ip-172-31-23-116:~/aws-fleet$ ansible all -i inventory.ini -a "uptime"
54.204.128.25 | CHANGED | rc=0 >>
 10:01:59 up 57 min,  1 user,  load average: 0.00, 0.00, 0.00
50.17.60.116 | CHANGED | rc=0 >>
 10:01:59 up 57 min,  1 user,  load average: 0.00, 0.00, 0.00
52.91.23.50 | CHANGED | rc=0 >>
 10:01:59 up 57 min,  1 user,  load average: 0.00, 0.00, 0.00
35.172.183.242 | CHANGED | rc=0 >>
 10:01:59 up 57 min,  1 user,  load average: 0.00, 0.00, 0.00
ubuntu@ip-172-31-23-116:~/aws-fleet$
```

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PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Install Nginx (on web servers)

ansible web -i inventory.ini -b -m apt -a "name=nginx state=present update_cache=yes"



Ensure Nginx is running

```
ansible web -i inventory.ini -b -m service -a "name=nginx state=started enabled=yes"
```

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Disk usage

ansible all -i inventory.ini -a "df -h"

```
aws [Search] [Alt+S] United States (N. Virginia) Account ID: 3111-4154-2113 Adrina Colaco
```

```
ubuntu@ip-172-31-23-116:~/aws-fleet$ ansible all -i inventory.ini -a "df -h"
35.172.183.242 | CHANGED | rc=0 >>
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        7.6G  1.9G  5.7G  26% /
tmpfs            475M   0  475M   0% /dev/shm
tmpfs           190M  848K  190M   1% /run
tmpfs            5.0M   0   5.0M   0% /run/lock
/dev/xvda15     105M  6.1M   99M   6% /boot/efi
tmpfs           95M  4.0K   95M   1% /run/user/1000
50.17.60.116 | CHANGED | rc=0 >>
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        7.6G  2.9G  5.7G  26% /
tmpfs            475M   0  475M   0% /dev/shm
tmpfs           190M  852K  190M   1% /run
tmpfs            5.0M   0   5.0M   0% /run/lock
/dev/xvda15     105M  6.1M   99M   6% /boot/efi
tmpfs           95M  4.0K   95M   1% /run/user/1000
54.204.128.25 | CHANGED | rc=0 >>
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        7.6G  1.9G  5.8G  24% /
tmpfs            475M   0  475M   0% /dev/shm
tmpfs           190M  848K  190M   1% /run
tmpfs            5.0M   0   5.0M   0% /run/lock
/dev/xvda15     105M  6.1M   99M   6% /boot/efi
tmpfs           95M  4.0K   95M   1% /run/user/1000
52.91.23.50 | CHANGED | rc=0 >>
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        7.6G  2.9G  5.7G  26% /
tmpfs            475M   0  475M   0% /dev/shm
tmpfs           190M  852K  190M   1% /run
tmpfs            5.0M   0   5.0M   0% /run/lock
/dev/xvda15     105M  6.1M   99M   6% /boot/efi
tmpfs           95M  4.0K   95M   1% /run/user/1000
ubuntu@ip-172-31-23-116:~/aws-fleet$

i-082f0d8b3ad04ba3b (LinuxAdrina)
PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116
```

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Memory usage

ansible all -i inventory.ini -a "free -m"

```
ubuntu@ip-172-31-23-116:~/aws-fleet$ ansible all -i inventory.ini -a "free -m"
50.17.60.116 | CHANGED | rc=0 >>
              total        used         free      shared  buff/cache   available
Mem:          949          176           321           0           452           611
Swap:           0             0             0
52.91.23.50 | CHANGED | rc=0 >>
              total        used         free      shared  buff/cache   available
Mem:          949          175           298           0           475           611
Swap:           0             0             0
35.172.183.242 | CHANGED | rc=0 >>
              total        used         free      shared  buff/cache   available
Mem:          949          172           196           0           580           616
Swap:           0             0             0
54.204.128.25 | CHANGED | rc=0 >>
              total        used         free      shared  buff/cache   available
Mem:          949          174           207           0           567           615
Swap:           0             0             0
ubuntu@ip-172-31-23-116:~/aws-fleet$
```

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PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Assignment 35 – Multi-Play Web Deploy on Azure (Learn-How to read module)

Project Setup

Folder structure

```
static-web/  
├─ inventory.ini  
├─ site.yml  
├─ files/  
│   └─ index.html  
└─ README.md
```

```
ubuntu@ip-172-31-23-116:~/static-web$ ls -l  
total 12  
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 31 10:55 files  
-rw-rw-r-- 1 ubuntu ubuntu  110 Oct 31 10:58 inventory.ini  
-rw-rw-r-- 1 ubuntu ubuntu 1066 Oct 31 11:04 site.yml  
ubuntu@ip-172-31-23-116:~/static-web$ cd files/  
ubuntu@ip-172-31-23-116:~/static-web/files$ ls -l  
total 16  
-rw-rw-r-- 1 ubuntu ubuntu 15814 Oct 31 10:55 index.html  
ubuntu@ip-172-31-23-116:~/static-web/files$
```

1 Step 1: Download the web content

From <https://github.com/pravinmishraaws/Azure-Static-Website>:

```
mkdir -p static-web/files
```

```
cd static-web/files
```

```
wget https://raw.githubusercontent.com/pravinmishraaws/Azure-Static-Website/main/index.html
```

```
ubuntu@ip-172-31-23-116:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-23-116:~$ mkdir -p static-web/files
ubuntu@ip-172-31-23-116:~$ cd static-web/files
ubuntu@ip-172-31-23-116:~/static-web/files$ wget https://raw.githubusercontent.com/pravinmishraaws/Azure-Static-Website/main/index.html
--2025-10-31 10:55:02-- https://raw.githubusercontent.com/pravinmishraaws/Azure-Static-Website/main/index.html
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.110.133, 185.199.111.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 15814 (15K) [text/plain]
Saving to: 'index.html'

index.html                               100%[=====] 15.44K --.-KB/s  in 0s

2025-10-31 10:55:02 (126 MB/s) - 'index.html' saved [15814/15814]
ubuntu@ip-172-31-23-116:~/static-web/files$
```

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PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

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🌿 Step 2: Create your Ansible inventory

inventory.ini

[web]

<your_web1_public_ip>

<your_web2_public_ip>

[all:vars]

ansible_user=ubuntu

ansible_ssh_private_key_file=~/.ssh/id_ed25519

```
GNU nano 7.2                                inventory.ini
[web]
50.17.60.116
52.91.23.50

[all:vars]
ansible_user=ubuntu
ansible_ssh_private_key_file=~/.ssh/id_ed25519
```

[Wrote 7 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Un
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line	M-E Re

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Step 3: Create the multi-play playbook

site.yml

```
---
# Play 1: Prepare web servers
- name: Install and configure web server
  hosts: web
  become: yes
  tasks:
    - name: Update apt cache
      apt:
        update_cache: yes

    - name: Install nginx
      apt:
        name: nginx
        state: present
```

- name: Ensure nginx is running and enabled

service:

- name: nginx
- state: started
- enabled: yes

Play 2: Deploy static content

- name: Deploy static website files

hosts: web

become: yes

tasks:

- name: Copy index.html to nginx web root

copy:

- src: files/index.html
- dest: /var/www/html/index.html
- owner: www-data
- group: www-data
- mode: '0644'

Play 3: Verify deployment

- name: Verify website availability

hosts: localhost

gather_facts: no

vars:

- target_web: "{{ groups['web'] | first }}"

tasks:

- name: Check website using URI module

uri:

- url: "http://{{ target_web }}"
- return_content: yes
- register: result

- name: Display result

debug:

- var: result.content

```
GNU nano 7.2 site.yml *
--
# Play 1: Prepare web servers
- name: Install and configure web server
  hosts: web
  become: yes
  tasks:
    - name: Update apt cache
      apt:
        update_cache: yes

    - name: Install nginx
      apt:
        name: nginx
        state: present

    - name: Ensure nginx is running and enabled
      service:
        name: nginx
        state: started
        enabled: yes

# Play 2: Deploy static content
- name: Deploy static website files
  hosts: web
  become: yes
  tasks:
    - name: Copy index.html to nginx web root
      copy:

^G Help      ^O Write Out  ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File  ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line  M-E Redo
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Step 4: Run the playbook

From inside static-web/:

```
ansible-playbook -i inventory.ini site.yml
```

You should see:

- Nginx installed and started on web hosts
- index.html copied successfully
- URI check showing HTML content in the output

```
PLAY [Install and configure web server] *****
```

```

#####
#ACSV: I'll update ant cache! #####

```

```
#####
TAPSV [install again] #####
```

```

#####
# ORC resource name is missing and enabled #####
#####

```

```

OK: [20,17,00,110]
=====
mean fm 3      1.1      1.1      613 3
=====

```

```
Task: [Gathering Facts]
ok: [52.91.23.50]
```

```
OK: [50,17,00,110]
```

```
changed: [30.17.00.110]
```

```
TASK [Check website using HPI module] *****
```

```
TASK [Display result] *****
ok: [localhost] => {
```

1. *Journal of Management Studies*, 1996, 33, 1, 1-14.

```
ubuntu@ip-172-31-23-116:~/static-web$
```

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Step 5: Manual verification

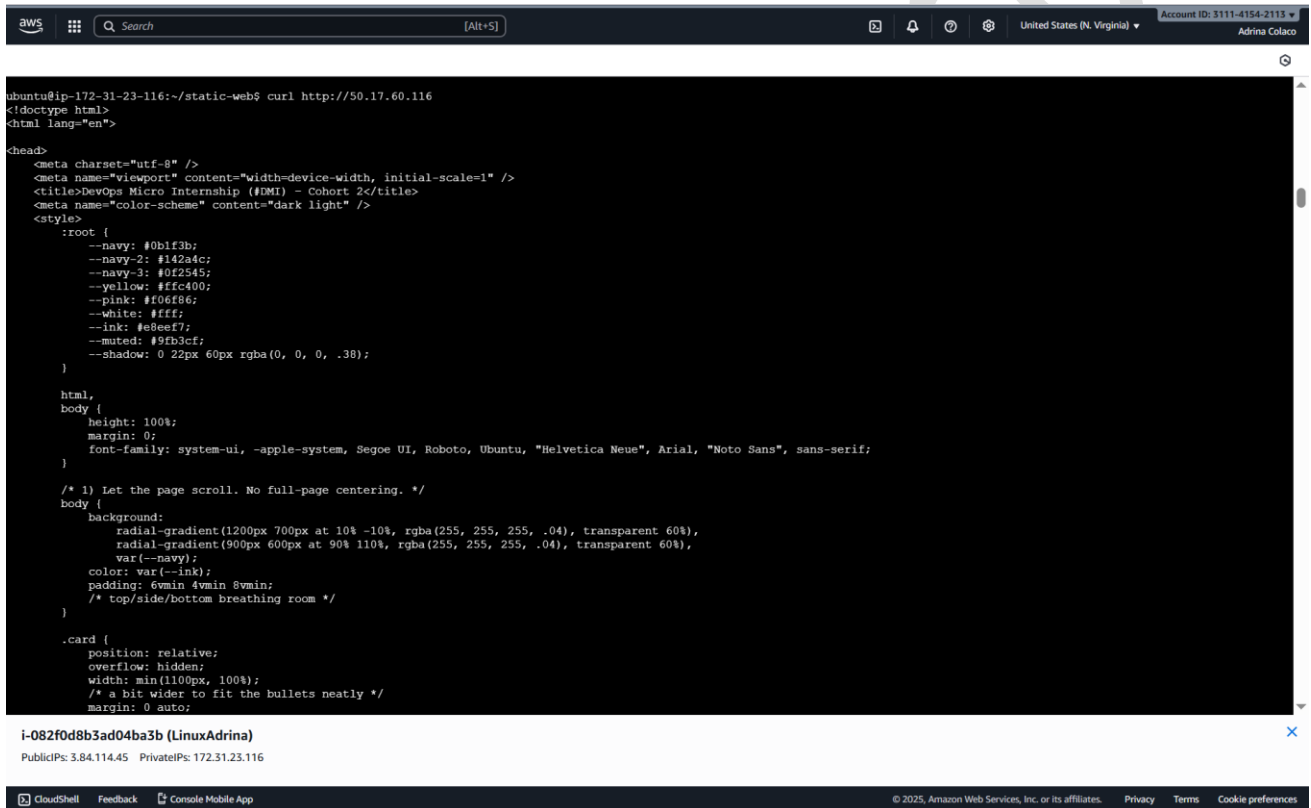
Open your browser or run:

```
curl http://<web_ip>
```

You should see the same HTML content as in

<https://ebstaticwebsite.z29.web.core.windows.net>.

```
curl http://50.17.60.116
```



```
ubuntu@ip-172-31-23-116:~/static-web$ curl http://50.17.60.116
<!doctype html>
<html lang="en">

<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>DevOps Micro Internship (#DMI) - Cohort 2</title>
  <meta name="color-scheme" content="dark light" />
  <style>
    :root {
      --navy: #0b1f3b;
      --navy-2: #142a4c;
      --navy-3: #0f2545;
      --yellow: #ffc400;
      --pink: #f06f96;
      --white: #fff;
      --ink: #8eef77;
      --muted: #9fb3cf;
      --shadow: 0 22px 60px rgba(0, 0, 0, .38);
    }

    html,
    body {
      height: 100%;
      margin: 0;
      font-family: system-ui, -apple-system, Segoe UI, Roboto, Ubuntu, "Helvetica Neue", Arial, "Noto Sans", sans-serif;
    }

    /* 1) Let the page scroll. No full-page centering. */
    body {
      background:
        radial-gradient(1200px 700px at 10% -10%, rgba(255, 255, 255, .04), transparent 60%),
        radial-gradient(900px 600px at 90% 110%, rgba(255, 255, 255, .04), transparent 60%),
        var(--navy);
      color: var(--ink);
      padding: 6vmin 4vmin 8vmin;
      /* top/side/bottom breathing room */
    }

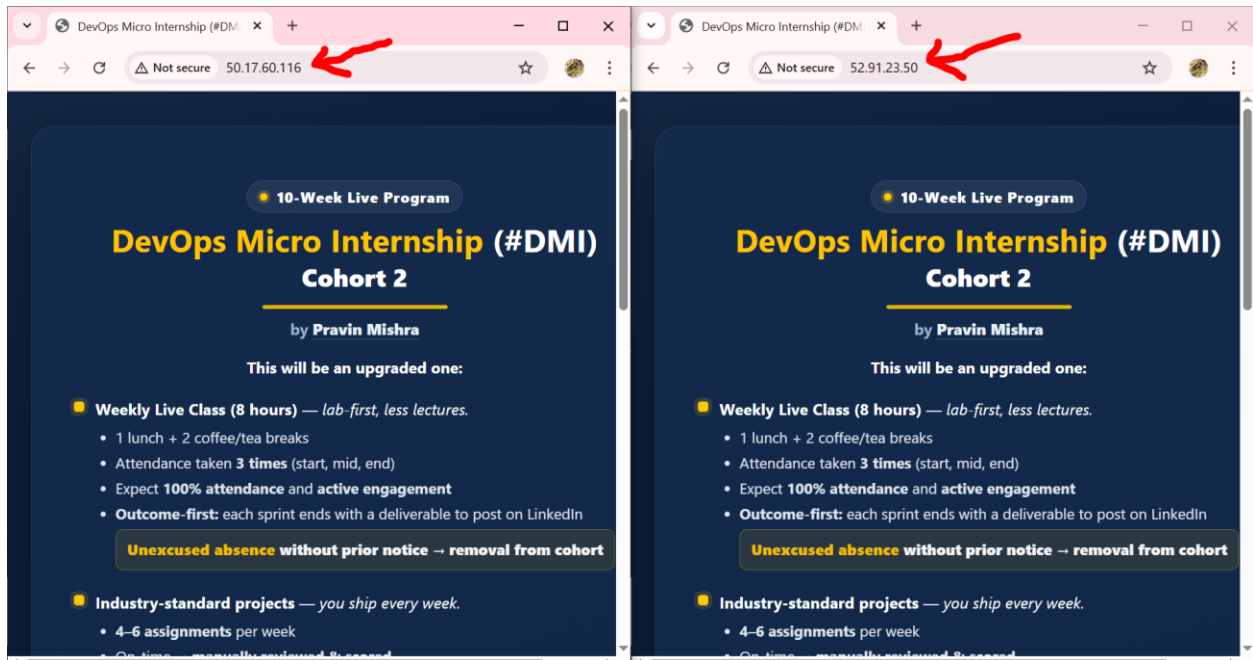
    .card {
      position: relative;
      overflow: hidden;
      width: min(1100px, 100%);
      /* a bit wider to fit the bullets neatly */
      margin: 0 auto;
    }

i-082f0d8b3ad04ba3b (LinuxAdrina)
PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116
```

```
curl http://52.91.23.50
```




Browser Verification



Assignment 36 – Deploy Mini Finance Project using Terraform and Ansible

 Folder Layout

```
mini-finance/  
├─ terraform/  
│   ├── providers.tf  
│   ├── main.tf  
│   ├── variables.tf  
│   └─ outputs.tf  
├─ ansible/  
│   ├── inventory.ini  
│   └─ site.yml  
└─ README.md
```

- terraform/ has all infra-as-code
- ansible/ has inventory & playbook
- README.md explains how to run & what to submit

```

ubuntu@ip-172-31-23-116:~/mini-finance$ ls -l
total 12
-rw-rw-r-- 1 ubuntu ubuntu 855 Oct 31 15:15 README.md
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 31 14:29 ansible
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 31 12:20 terraform
ubuntu@ip-172-31-23-116:~/mini-finance$ cd terraform/
ubuntu@ip-172-31-23-116:~/mini-finance/terraform$ ls -l
total 36
-rw-rw-r-- 1 ubuntu ubuntu 2734 Oct 31 12:09 main.tf
-rw-rw-r-- 1 ubuntu ubuntu 118 Oct 31 12:15 outputs.tf
-rw-rw-r-- 1 ubuntu ubuntu 203 Oct 31 12:04 providers.tf
-rw-rw-r-- 1 ubuntu ubuntu 19148 Oct 31 12:20 terraform.tfstate
-rw-rw-r-- 1 ubuntu ubuntu 504 Oct 31 12:06 variables.tf
ubuntu@ip-172-31-23-116:~/mini-finance/terraform$ cd ..
ubuntu@ip-172-31-23-116:~/mini-finance$ cd ansible/
ubuntu@ip-172-31-23-116:~/mini-finance/ansible$ ls -l
total 8
-rw-rw-r-- 1 ubuntu ubuntu 99 Oct 31 13:32 inventory.ini
-rw-rw-r-- 1 ubuntu ubuntu 1830 Oct 31 14:19 site.yml
ubuntu@ip-172-31-23-116:~/mini-finance/ansible$

```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

1. Terraform: Provision AWS VM + NSG

In terraform/, create the following files:

providers.tf

```

terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

```

```

provider "aws" {
  region = var.aws_region
  profile = var.aws_profile # optional
}

```

```
GNU nano 7.2 providers.tf
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = var.aws_region
  profile = var.aws_profile # optional
}
```

variables.tf (optional but good for reuse)

```
variable "aws_region" {
  description = "AWS region to use"
  type        = string
  default     = "us-east-1"
}
```

```
variable "aws_profile" {
  description = "AWS CLI profile (optional)"
  type        = string
  default     = "default"
}
```

```
variable "instance_name" {
  description = "Name tag for the EC2 instance"
  type        = string
  default     = "mini-finance-server"
}
```

```
variable "ssh_key_name" {
  description = "Name of the AWS Key Pair"
  type        = string
  default     = "mini-finance-key"
}
```

```
GNU nano 7.2 variables.tf *
variable "aws_region" {
  description = "AWS region to use"
  type        = string
  default     = "us-east-1"
}

variable "aws_profile" {
  description = "AWS CLI profile (optional)"
  type        = string
  default     = "default"
}

variable "instance_name" {
  description = "Name tag for the EC2 instance"
  type        = string
  default     = "mini-finance-server"
}

variable "ssh_key_name" {
  description = "Name of the AWS Key Pair"
  type        = string
  default     = "mini-finance-key"
}
```

main.tf

VPC

```
resource "aws_vpc" "main" {
  cidr_block = "10.0.0.0/16"
  tags = {
    Name = "mini-finance-vpc"
  }
}
```

Subnet

```
resource "aws_subnet" "public" {
  vpc_id      = aws_vpc.main.id
  cidr_block  = "10.0.1.0/24"
  map_public_ip_on_launch = true
  tags = {
    Name = "mini-finance-public-subnet"
  }
}
```

Internet Gateway

```
resource "aws_internet_gateway" "igw" {
  vpc_id = aws_vpc.main.id
  tags = {
    Name = "mini-finance-igw"
  }
}
```

```
}  
}
```

Route Table & Route

```
resource "aws_route_table" "rt" {  
  vpc_id = aws_vpc.main.id  
  route {  
    cidr_block = "0.0.0.0/0"  
    gateway_id = aws_internet_gateway.igw.id  
  }  
  tags = {  
    Name = "mini-finance-rt"  
  }  
}
```

```
resource "aws_route_table_association" "rt_assoc" {  
  subnet_id    = aws_subnet.public.id  
  route_table_id = aws_route_table.rt.id  
}
```

Security Group (for SSH + HTTP)

```
resource "aws_security_group" "web_sg" {  
  name      = "mini-finance-sg"  
  description = "Allow SSH and HTTP access"  
  vpc_id    = aws_vpc.main.id
```

```
  ingress {  
    description = "SSH"  
    from_port   = 22  
    to_port     = 22  
    protocol    = "tcp"  
    cidr_blocks = ["0.0.0.0/0"]  
  }  
}
```

```
  ingress {  
    description = "HTTP"  
    from_port   = 80  
    to_port     = 80
```

```
    protocol = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
}

egress {
    from_port = 0
    to_port = 0
    protocol = "-1"
    cidr_blocks = ["0.0.0.0/0"]
}

tags = {
    Name = "mini-finance-sg"
}
}

# Key pair
resource "aws_key_pair" "app_key" {
    key_name = var.ssh_key_name
    public_key = file("~/ssh/id_ed25519.pub")
}

# EC2 Instance
resource "aws_instance" "web" {
    ami = data.aws_ami.ubuntu.id
    instance_type = "t3.micro"
    subnet_id = aws_subnet.public.id
    vpc_security_group_ids = [ aws_security_group.web_sg.id ]
    key_name = aws_key_pair.app_key.key_name

    tags = {
        Name = var.instance_name
    }
}

# To use public IP
associate_public_ip_address = true

# Optional: install basic tools for Ansible connectivity
```



```
provisioner "remote-exec" {
  connection {
    type      = "ssh"
    user      = "ubuntu"
    private_key = file("~/ssh/id_ed25519")
    host      = self.public_ip
  }

  inline = [
    "sudo apt update -y",
    "sudo apt install -y python3"
  ]
}

# Data source for Amazon Linux / Ubuntu AMI
data "aws_ami" "ubuntu" {
  most_recent = true
  owners      = ["099720109477"] # Canonical
  filter {
    name = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-focal-*-amd64-server-*"]
  }
}

# Outputs
output "public_ip" {
  description = "Public IP of the web server"
  value       = aws_instance.web.public_ip
}

output "ssh_key_name" {
  value = aws_key_pair.app_key.key_name
}
```

AWS

Q Search

[Alt+S]

United States (N. Virginia)

Account ID: 3111-4154-2113

Adrina Colaco

GNU nano 7.2

main.tf

```
resource "aws_vpc" "main" {
  cidr_block = "10.0.0.0/16"
  tags = {
    Name = "mini-finance-vpc"
  }
}

# Subnet
resource "aws_subnet" "public" {
  vpc_id = aws_vpc.main.id
  cidr_block = "10.0.1.0/24"
  map_public_ip_on_launch = true
  tags = {
    Name = "mini-finance-public-subnet"
  }
}

# Internet Gateway
resource "aws_internet_gateway" "igw" {
  vpc_id = aws_vpc.main.id
  tags = {
    Name = "mini-finance-igw"
  }
}

# Route Table & Route
resource "aws_route_table" "rt" {
  vpc_id = aws_vpc.main.id
  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.igw.id
  }
  tags = {
    Name = "mini-finance-rt"
  }
}

resource "aws_route_table_association" "rt_assoc" {
  subnet_id = aws_subnet.public.id
  route_table_id = aws_route_table.rt.id
}

# Security Group (for SSH + HTTP)
resource "aws_security_group" "web_sg" {
  name = "mini-finance-sg"
  description = "Allow SSH and HTTP access"
  vpc_id = aws_vpc.main.id

  ingress {
    description = "SSH"
    from_port = 22
    to_port = 22
    protocol = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    description = "HTTP"
    from_port = 80
    to_port = 80
    protocol = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port = 0
    to_port = 0
    protocol = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

  tags = {
    Name = "mini-finance-sg"
  }
}

# Key pair
resource "aws_key_pair" "app_key" {
  key_name = var.ssh_key_name
  public_key = file("~/ssh/id_ed25519.pub")
}

# EC2 Instance
resource "aws_instance" "web" {
  ami = data.aws_ami.ubuntu.id
  instance_type = "t3.micro"
  subnet_id = aws_subnet.public.id
  vpc_security_group_ids = [aws_security_group.web_sg.id]
  key_name = aws_key_pair.app_key.key_name

  tags = {
    Name = var.instance_name
  }

  # To use public IP
  associate_public_ip_address = true

  # Optional: install basic tools for Ansible connectivity
  provisioner "remote-exec" {
    connection {
      type = "ssh"
      user = "ubuntu"
      private_key = file("~/ssh/id_ed25519")
      host = self.public_ip
    }
  }
}
```

Help

Write Out

Where Is

Cut

Execute

Location

Undo

Set Mark

To Bracket

Previous

Back

Exit

Read File

Replace

Paste

Justify

Go To Line

Redo

Copy

Where Was

Next

Forward

i-082f0d8b3ad04ba3b (LinuxAdrina)

X

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

CloudShell

Feedback

Console Mobile App

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outputs.tf

```
output "web_public_ip" {  
  
    description = "Public IPv4 of the web server"  
  
    value      = aws_instance.web.public_ip  
  
}
```

```
GNU nano 7.2                                     outputs.tf  
output "web_public_ip" {  
    description = "Public IPv4 of the web server"  
    value      = aws_instance.web.public_ip  
}
```

README.md

```
GNU nano 7.2                                     README.md  
Project Overview:  
The Mini Finance project is a static website that helps users manage their financial data. The key sections include:  
  
✔ Dashboard Overview - Displays account balance, transactions, and financial insights.  
✔ Profile Page - Stores user details like name, email, and contact information.  
✔ My Wallet - Allows users to manage payment methods.  
✔ Settings & Help Center - Enables users to update preferences and seek assistance.  
  
What you will do:  
◆ Cloning a Repository - Work with remote repositories on GitHub.  
◆ Branching in Git - Create and switch between feature branches.  
◆ Staging & Committing Changes - Track modifications properly.  
◆ Pushing Changes to GitHub - Synchronize your work with the remote repository.  
◆ Submitting a Pull Request (PR) - Collaborate using industry best practices.
```

2. Deploy Infrastructure with Terraform

```
cd mini-finance/terraform
```

```
terraform init
```

```
ubuntu@ip-172-31-23-116:~/mini-finance/terraform$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

terraform plan

terraform apply -auto-approve

Adriana Colaco



After apply, note down the public_ip output. Ensure the security group allows inbound 22 and 80.

```
Apply complete! Resources: 8 added, 0 changed, 0 destroyed.
```

```
Outputs:
```

```
public_ip = "3.219.170.155"
ssh_key_name = "mini-finance-key"
web_public_ip = "3.219.170.155"
ubuntu@ip-172-31-23-116:~/mini-finance/terraform$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

3. Passwordless SSH Check

Test SSH:

```
ssh -i ~/.ssh/id_ed25519 ubuntu@<public_ip>
```

hostname

If that works passwordlessly, infra is ready.

The screenshot shows an AWS CloudShell terminal window. At the top, the AWS logo and search bar are visible. The terminal output shows the user running a series of commands to test a passwordless SSH connection to an Ubuntu instance. The commands and their outputs are as follows:

```
ubuntu@ip-172-31-23-116:~$ cd mini-finance/
ubuntu@ip-172-31-23-116:~/mini-finance$ ssh -i ~/.ssh/id_ed25519 ubuntu@3.219.170.155
The authenticity of host '3.219.170.155 (3.219.170.155)' can't be established.
ED25519 key fingerprint is SHA256:hwacJvQGVfj6IZjy16ymm20rzDdIVQ2L3qNFWac11CY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.219.170.155' (ED25519) to the list of known hosts.
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1084-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Oct 31 13:30:17 UTC 2025

System load: 0.0          Processes: 102
Usage of /: 24.8% of 7.57GB Users logged in: 0
Memory usage: 22%        IPV4 address for ens5: 10.0.1.92
Swap usage: 0%

Expanded Security Maintenance for Infrastructure is not enabled.

15 updates can be applied immediately.
15 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

44 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Oct 31 12:20:36 2025 from 3.84.114.45
ubuntu@ip-10-0-1-92:~$ exit
logout
Connection to 3.219.170.155 closed.
ubuntu@ip-172-31-23-116:~/mini-finance$
```

Below the terminal output, the instance details are shown:

i-082f0d8b3ad04ba3b (LinuxAdrina)
PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

The bottom of the screenshot shows the AWS CloudShell footer with the CloudShell logo, Feedback link, Console Mobile App link, and copyright information: © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences.

4. Ansible: Multi-Play Playbook + Inventory

In mini-finance/ansible/ create:

inventory.ini

[web]

<public_ip>

[all:vars]

ansible_user=ubuntu

ansible_ssh_private_key_file=~/.ssh/id_ed25519

```
GNU nano 7.2 inventory.ini
[web]
3.219.170.155

[all:vars]
ansible_user=ubuntu
ansible_ssh_private_key_file=~/.ssh/id_ed25519
```

site.yml

This playbook performs multiple plays:

Play 1: install & configure Nginx

- name: "Provision web server with Nginx"

hosts: web

become: yes

tasks:

- name: Update apt cache

apt:

update_cache: yes

cache_valid_time: 3600

- name: Install Nginx

apt:

name: nginx

state: present

- name: Enable and start Nginx

service:

name: nginx

enabled: yes

state: started

Play 2: deploy mini-finance static site

- name: Deploy mini-finance static site (clone on remote then sync using synchronize)

hosts: web

become: yes

tasks:

- name: Clone the static mini-finance repo on remote

git:

repo: "https://github.com/pravinmishraaws/MiniFinanceDemo.git"

dest: /tmp/mini-finance-site/

version: main

- name: Sync site contents from the cloned folder to nginx web root

ansible.posix.synchronize:

src: /tmp/mini-finance-site/

dest: /var/www/html/

recursive: yes

delegate_to: "{{ inventory_hostname }}"

- name: Set ownership / permissions of files in /var/www/html/

file:

path: /var/www/html/

state: directory

recurse: yes

owner: www-data

group: www-data

mode: "0755"

Play 3: reload nginx gracefully

- name: "Reload Nginx after deployment"

hosts: web

become: yes

tasks:

- name: Reload Nginx

service:

name: nginx

state: reloaded

Play 4: verify HTTP success

- name: "Verify site is up via HTTP"

hosts: localhost

gather_facts: no

vars:

target: "{{ groups['web'] | first }}"

tasks:

- name: Check HTTP status

uri:

url: "http://{{ target }}"

return_content: false

status_code: 200

register: result

- name: Debug result

debug:

var: result.status

```
GNU nano 7.2 site.yml
# Play 1: Install & configure Nginx
- name: "Provision web server with Nginx"
  hosts: web
  become: yes
  tasks:
    - name: Update apt cache
      apt:
        update_cache: yes
        cache_valid_time: 3600

    - name: Install Nginx
      apt:
        name: nginx
        state: present

    - name: Enable and start Nginx
      service:
        name: nginx
        enabled: yes
        state: started

# Play 2: Deploy mini-finance static site
- name: Deploy mini-finance static site (clone on remote then sync using synchronize)
  hosts: web
  become: yes

  tasks:
    - name: Clone the static mini-finance repo on remote
      git:
        repo: "https://github.com/pravinmishraaws/mini_finance.git"
        dest: /tmp/mini-finance-site/
        version: main

    - name: Sync site contents from the cloned folder to nginx web root
      ansible.posix.synchronize:
        src: /tmp/mini-finance-site/
        dest: /var/www/html/
        recursive: yes
        delegate_to: "{{ inventory_hostname }}"

    - name: Set ownership / permissions of files in /var/www/html/
      file:
        path: /var/www/html/
        state: directory
        recurse: yes
        owner: www-data
        group: www-data
        mode: "0755"

# Play 3: reload nginx gracefully
- name: "Reload Nginx after deployment"
  hosts: web
  become: yes
  tasks:
    - name: Reload Nginx
      service:
        name: nginx
        state: reloaded

# Play 4: verify HTTP success
```

i-082f0d8b3ad04ba3b (LinuxAdrina)
PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

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5. Run Ansible Playbook

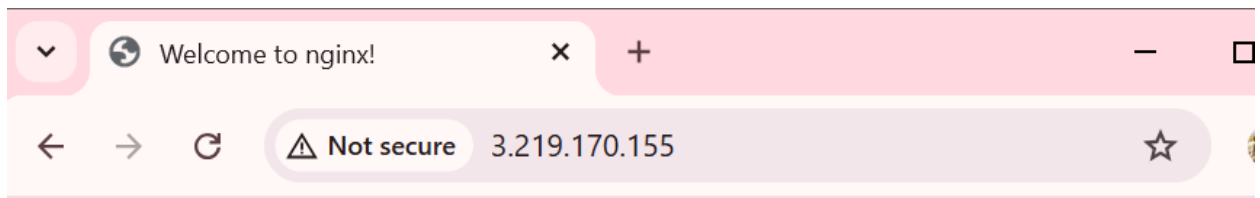
From mini-finance/:

cd ansible

ansible-playbook -i inventory.ini site.yml

Check that:

- Nginx installed, running



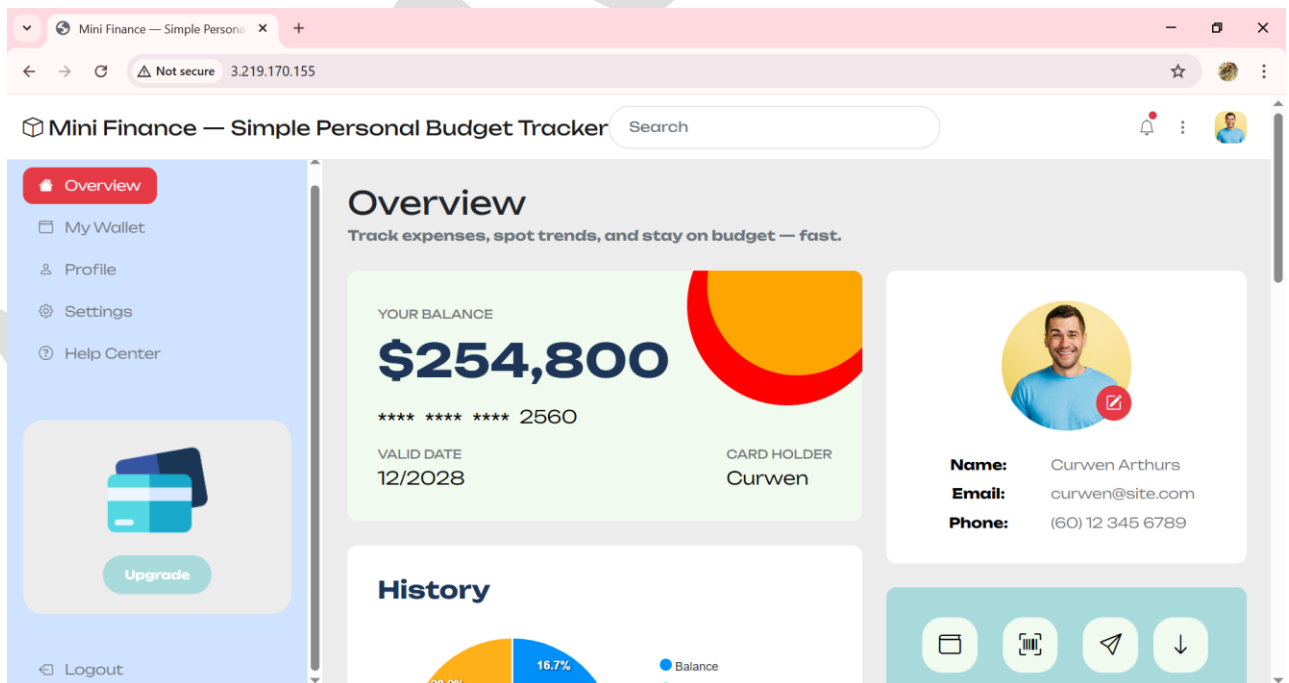
Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

- The site renders properly (open browser `http://<public_ip>`)



- Playbook shows verification succeeded

```
ubuntu@ip-172-31-23-116:~/mini-finance/ansible$ ansible-playbook -i inventory.ini site.yml

PLAY ["Provision web server with Nginx"] *****
TASK [Gathering Facts] *****
ok: [3.219.170.155]

TASK [Update apt cache] *****
ok: [3.219.170.155]

TASK [Install Nginx] *****
ok: [3.219.170.155]

TASK [Enable and start Nginx] *****
ok: [3.219.170.155]

PLAY [Deploy mini-finance static site (clone on remote then sync using synchronize)] *****
TASK [Gathering Facts] *****
ok: [3.219.170.155]

TASK [Clone the static mini-finance repo on remote] *****
changed: [3.219.170.155]

TASK [Sync site contents from the cloned folder to nginx web root] *****
changed: [3.219.170.155]

TASK [Set ownership / permissions of files in /var/www/html/] *****
changed: [3.219.170.155]

PLAY ["Reload Nginx after deployment"] *****
TASK [Gathering Facts] *****
ok: [3.219.170.155]

TASK [Reload Nginx] *****
changed: [3.219.170.155]

PLAY ["Verify site is up via HTTP"] *****
TASK [Check HTTP status] *****
ok: [localhost]

TASK [Debug result] *****
ok: [localhost] => {
  "result.status": "200"
}

PLAY RECAP *****
3.219.170.155      : ok=10   changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
localhost         : ok=2     changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu@ip-172-31-23-116:~/mini-finance/ansible$ nano site.yml
ubuntu@ip-172-31-23-116:~/mini-finance/ansible$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)
PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

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Assignment 37 – Production-Grade EpicBook — Terraform + Ansible Roles (Azure or AWS)

📁 Project Layout

```
epicbook-prod/
├─ terraform/
│   └─ aws/
│       ├── providers.tf
│       ├── main.tf
│       ├── variables.tf
│       └─ outputs.tf
├─ ansible/
│   ├── inventory.ini
│   ├── group_vars/
│   │   └─ web.yml
│   ├── roles/
│   │   ├── common/
│   │   │   ├── tasks/
│   │   │   │   └─ main.yml
│   │   │   └─ handlers/
│   │   │       └─ main.yml
│   │   ├── nginx/
│   │   │   ├── tasks/
│   │   │   │   └─ main.yml
│   │   │   ├── handlers/
│   │   │   │   └─ main.yml
│   │   │   └─ templates/
│   │   │       └─ epicbook.conf.j2
│   │   └─ epicbook/
│   │       ├── tasks/
│   │       │   └─ main.yml
│   │       └─ handlers/
│   │           └─ main.yml
│   └─ site.yml
└─ README.md
```

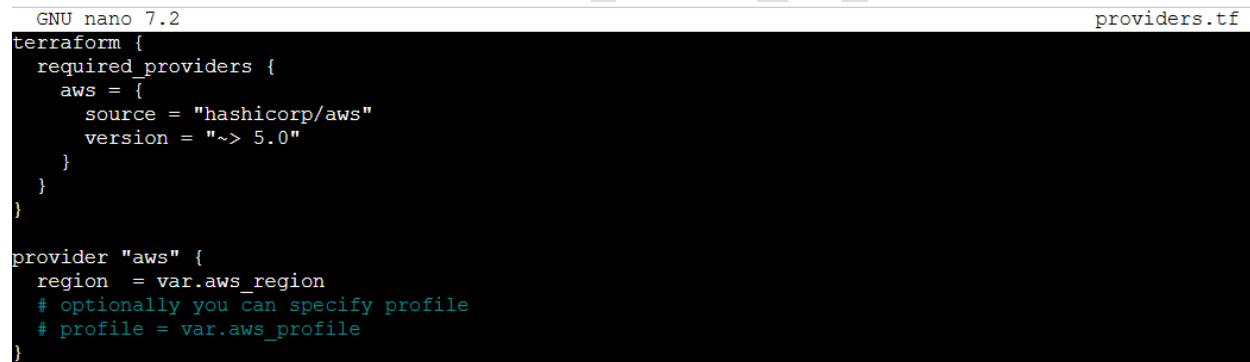
Step 1: Terraform to Provision an AWS VM

In terraform/aws/ do something like:

providers.tf

```
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = var.aws_region
  # optionally you can specify profile
  # profile = var.aws_profile
}
```



The screenshot shows a terminal window with the nano 7.2 text editor open to a file named providers.tf. The editor's interface includes a title bar with 'GNU nano 7.2' on the left and 'providers.tf' on the right. The code content is displayed on a dark background with syntax highlighting: 'terraform {' is in light blue, 'required_providers {' is in light green, 'aws = {' is in light blue, 'source = "hashicorp/aws"' is in light green, 'version = "~> 5.0"' is in light green, '}' is in light blue, and 'provider "aws" {' is in light blue. The comments '# optionally you can specify profile' and '# profile = var.aws_profile' are in light green. The code is as follows:

```
GNU nano 7.2 providers.tf
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = var.aws_region
  # optionally you can specify profile
  # profile = var.aws_profile
}
```

variables.tf

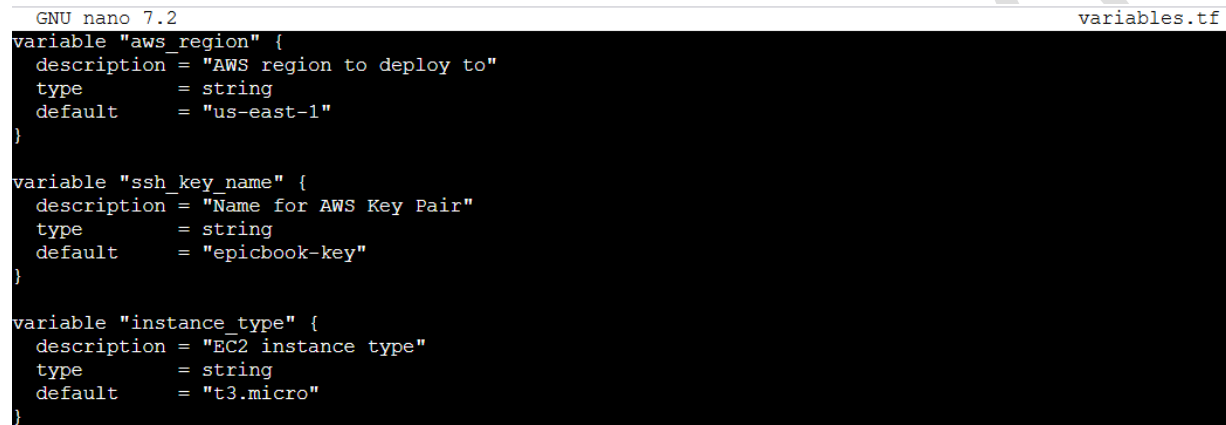
Define variables, e.g.:

```
variable "aws_region" {
  description = "AWS region to deploy to"
  type        = string
  default     = "us-east-1"
}
```

```
variable "ssh_key_name" {
  description = "Name for AWS Key Pair"
  type        = string
}
```

```
    default = "epicbook-key"
}

variable "instance_type" {
    description = "EC2 instance type"
    type        = string
    default     = "t3.micro"
}
```



```
GNU nano 7.2 variables.tf
variable "aws_region" {
    description = "AWS region to deploy to"
    type        = string
    default     = "us-east-1"
}

variable "ssh_key_name" {
    description = "Name for AWS Key Pair"
    type        = string
    default     = "epicbook-key"
}

variable "instance_type" {
    description = "EC2 instance type"
    type        = string
    default     = "t3.micro"
}
```

main.tf

Create a VPC / Subnet / Security Group / EC2 instance:

```
resource "aws_vpc" "main" {
    cidr_block = "10.0.0.0/16"
    tags = {
        Name = "epicbook-vpc"
    }
}

resource "aws_subnet" "public" {
    vpc_id            = aws_vpc.main.id
    cidr_block        = "10.0.1.0/24"
    map_public_ip_on_launch = true
    tags = {
        Name = "epicbook-subnet-public"
    }
}
```

```
resource "aws_internet_gateway" "igw" {  
  vpc_id = aws_vpc.main.id  
}
```

```
resource "aws_route_table" "rt" {  
  vpc_id = aws_vpc.main.id  
  
  route {  
    cidr_block = "0.0.0.0/0"  
    gateway_id = aws_internet_gateway.igw.id  
  }  
}
```

```
resource "aws_route_table_association" "a" {  
  subnet_id    = aws_subnet.public.id  
  route_table_id = aws_route_table.rt.id  
}
```

```
resource "aws_security_group" "web_sg" {  
  name      = "epicbook-sg"  
  description = "Allow SSH and HTTP"  
  vpc_id    = aws_vpc.main.id
```

```
  ingress {  
    description = "SSH"  
    from_port   = 22  
    to_port     = 22  
    protocol    = "tcp"  
    cidr_blocks = ["0.0.0.0/0"]  
  }
```

```
  ingress {  
    description = "HTTP"  
    from_port   = 80  
    to_port     = 80  
    protocol    = "tcp"  
    cidr_blocks = ["0.0.0.0/0"]  
  }
```

```
}

egress {
  from_port = 0
  to_port   = 0
  protocol  = "-1"
  cidr_blocks = ["0.0.0.0/0"]
}

tags = {
  Name = "epicbook-sg"
}

resource "aws_key_pair" "key" {
  key_name   = var.ssh_key_name
  public_key = file("~/ssh/id_ed25519.pub")
}

data "aws_ami" "ubuntu" {
  most_recent = true
  owners      = ["099720109477"]
  filter {
    name = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
  }

  filter {
    name = "architecture"
    values = ["x86_64"]
  }

  filter {
    name = "root-device-type"
    values = ["ebs"]
  }
}
```

```
resource "aws_instance" "web" {
  ami          = data.aws_ami.ubuntu.id
  instance_type = var.instance_type
  subnet_id    = aws_subnet.public.id
  vpc_security_group_ids = [aws_security_group.web_sg.id]
  key_name     = aws_key_pair.key.key_name

  associate_public_ip_address = true

  tags = {
    Name = "epicbook-server"
  }

  # optional remote-exec to install python for Ansible connectivity
  provisioner "remote-exec" {
    connection {
      host     = self.public_ip
      type     = "ssh"
      user     = "ubuntu"
      private_key = file("~/ssh/id_ed25519")
    }
    inline = [
      "sudo apt update -y",
      "sudo apt install -y python3 python3-apt"
    ]
  }
}

output "public_ip" {
  value     = aws_instance.web.public_ip
  description = "Public IP of EpicBook server"
}

output "admin_user" {
  value     = "ubuntu"
  description = "SSH username for the instance"
}
```

AWS

Search

[Alt+S]

United States (N. Virginia)

Account ID: 3111-4154-2113

Adrina Colaco

GNU nano 7.2

main.tf

```
resource "aws_vpc" "vpc" {
  cidr_block = "10.0.0.0/16"
  tags = {
    Name = "epicbook-vpc"
  }
}

resource "aws_subnet" "public" {
  vpc_id = aws_vpc.main.id
  cidr_block = "10.0.1.0/24"
  map_public_ip_on_launch = true
  tags = {
    Name = "epicbook-subnet-public"
  }
}

resource "aws_internet_gateway" "igw" {
  vpc_id = aws_vpc.main.id
}

resource "aws_route_table" "rt" {
  vpc_id = aws_vpc.main.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.igw.id
  }
}

resource "aws_route_table_association" "a" {
  subnet_id = aws_subnet.public.id
  route_table_id = aws_route_table.rt.id
}

resource "aws_security_group" "web_sg" {
  name = "epicbook-sg"
  description = "Allow SSH and HTTP"
  vpc_id = aws_vpc.main.id

  ingress {
    description = "SSH"
    from_port = 22
    to_port = 22
    protocol = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    description = "HTTP"
    from_port = 80
    to_port = 80
    protocol = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    description = "HTTPS"
    from_port = 8080
    to_port = 8080
    protocol = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port = 0
    to_port = 0
    protocol = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

  tags = {
    Name = "epicbook-sg"
  }
}

resource "aws_key_pair" "key" {
  key_name = var.ssh_key_name
  public_key = file("~/ssh/id_ed25519.pub")
}

data "aws_ami" "ubuntu" {
  most_recent = true
  owners = ["099720109477"]
  filter {
    name = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-focal-22.04-amd64-server-*"]
  }
}

resource "aws_instance" "web" {
  ami = data.aws_ami.ubuntu.id
  instance_type = var.instance_type
  subnet_id = aws_subnet.public.id
  vpc_security_group_ids = [aws_security_group.web_sg.id]
  key_name = aws_key_pair.key.key_name

  associate_public_ip_address = true

  tags = {
    Name = "epicbook-server"
  }

  # optional remote-exec to install python for Ansible connectivity
  provisioner "remote-exec" {
    connection {
      host = self.public_ip
      type = "ssh"
      user = "ubuntu"
      private_key = file("~/ssh/id_ed25519")
    }
    inline = [
      "sudo apt update -y",
      "sudo apt install -y python3 python3-apt"
    ]
  }
}

output "public_ip" {
  value = aws_instance.web.public_ip
  description = "Public IP of EpicBook server"
}

output "admin_user" {
  value = "ubuntu"
  description = "SSH username for the instance"
}
```

Help

Write Out

Where Is

Cut

Execute

Location

Undo

Set Mark

To Bracket

Previous

Back

Exit

Read File

Replace

Paste

Justify

Go To Line

Redo

Copy

Where Was

Next

Forward

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIP: 3.84.114.45 PrivateIP: 172.31.23.116

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outputs.tf

(Optional duplication or extra formatting)

```
output "web_public_ip" {  
  
    description = "Public IP"  
  
    value      = aws_instance.web.public_ip  
}
```

```
GNU nano 7.2 outputs.tf  
output "web_public_ip" {  
    description = "Public IP"  
    value      = aws_instance.web.public_ip  
}
```

Step 2: Run Terraform

```
cd epicbook-prod/terraform/aws/
```

```
terraform init
```

```
ubuntu@ip-172-31-23-116:~/epicbook-prod/terraform/aws$ terraform init  
Initializing the backend...  
Initializing provider plugins...  
- Finding hashicorp/aws versions matching "~> 5.0"...  
- Installing hashicorp/aws v5.100.0...  
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)  
Terraform has created a lock file .terraform.lock.hcl to record the provider  
selections it made above. Include this file in your version control repository  
so that Terraform can guarantee to make the same selections by default when  
you run "terraform init" in the future.
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

```
ubuntu@ip-172-31-23-116:~/epicbook-prod/terraform/aws$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

terraform plan

Adriana Colaco

terraform apply -auto-approve

Adriana Colaco

The screenshot displays a Windows 7 desktop environment. A Notepad++ application window is open, showing a text file named 'C:\Users\user\Desktop\ipmac.txt'. The file contains a large list of IP addresses and their corresponding MAC addresses, organized into columns. The text is formatted with tabs and spaces, making it easy to read. The Notepad++ window is titled 'C:\Users\user\Desktop\ipmac.txt' and is open in 'Normal' view. The desktop background is a dark blue Windows 7 logo. The taskbar at the bottom shows the Start button, a few pinned applications, and the system tray with the clock showing 10:10 AM on 10/10/2015.

Then:

terraform output public_ip

```
Apply complete! Resources: 8 added, 0 changed, 0 destroyed.

Outputs:

admin_user = "ubuntu"
public_ip = "54.242.56.32"
web_public_ip = "54.242.56.32"
ubuntu@ip-172-31-23-116:~/epicbook-prod/terraform/aws$ terraform output public_ip
"54.242.56.32"
ubuntu@ip-172-31-23-116:~/epicbook-prod/terraform/aws$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Note the IP. Test SSH:

```
ssh -i ~/.ssh/id_ed25519 ubuntu@<public_ip> hostname
```

It should respond with the machine's hostname. This verifies SSH key & connectivity.

```
ubuntu@ip-172-31-23-116:~/epicbook-prod/terraform/aws$ ssh -i ~/.ssh/id_ed25519 ubuntu@54.242.56.32 hostname
The authenticity of host '54.242.56.32 (54.242.56.32)' can't be established.
ED25519 key fingerprint is SHA256:ZxVuZP0Cg7Zf8rqHBkf0pA4L9KPRghR4Ozj68yoA5T8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.242.56.32' (ED25519) to the list of known hosts.
ip-10-0-1-242
ubuntu@ip-172-31-23-116:~/epicbook-prod/terraform/aws$ ssh -i ~/.ssh/id_ed25519 ubuntu@54.242.56.32 hostname
ip-10-0-1-242
ubuntu@ip-172-31-23-116:~/epicbook-prod/terraform/aws$
```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

Step 3: Ansible Setup (Roles & Playbook)

In epicbook-prod/ansible/:

inventory.ini

[web]

<public_ip>

[all:vars]

ansible_user=ubuntu

ansible_ssh_private_key_file=~/.ssh/id_ed25519

```
GNU nano 7.2 inventory.ini
[web]
54.242.56.32

[all:vars]
ansible_user=ubuntu
ansible_ssh_private_key_file=~/.ssh/id_ed25519
```

group_vars/web.yml

You can set variables, e.g.:

group_vars/web.yml

epicbook_repo_url: "https://github.com/pravinmishraaws/theepicbook.git"

epicbook_dest: "/var/www/epicbook"

```
GNU nano 7.2 web.yml
---
# group_vars/web.yml
epicbook_repo_url: "https://github.com/pravinmishraaws/theepicbook.git"
epicbook_dest: "/var/www/epicbook"
```

site.yml

- name: Baseline / Common configuration

hosts: web

become: yes

roles:

- common

- name: Setup Nginx

hosts: web

become: yes

roles:

- nginx

```
- name: Deploy EpicBook app
  hosts: web
  become: yes
  roles:
    - epicbook
```

```
GNU nano 7.2 site.yml
---
- name: Baseline / Common configuration
  hosts: web
  become: yes
  roles:
    - common

- name: Setup Nginx
  hosts: web
  become: yes
  roles:
    - nginx

- name: Deploy EpicBook app
  hosts: web
  become: yes
  roles:
    - epicbook
```

Step 4: Define Ansible Roles

Role: common

ansible/roles/common/tasks/main.yml:

```
---
- name: Update apt cache
  apt:
    update_cache: yes

- name: Upgrade all packages
  apt:
    upgrade: dist
  register: upgrade_result

- name: Install common packages
  apt:
    name:
      - git
```

- curl
- ufw
- build-essential

state: present

update_cache: yes

- name: Ensure UFW allows SSH

ufw:

rule: allow

port: 22

proto: tcp

- name: Ensure UFW allows HTTP

ufw:

rule: allow

port: 80

proto: tcp

- name: Enable UFW

ufw:

state: enabled

- name: Disable SSH password authentication

lineinfile:

path: /etc/ssh/sshd_config

regexp: '^PasswordAuthentication'

line: 'PasswordAuthentication no'

backup: yes

notify: restart ssh

- name: Disable root login

lineinfile:

path: /etc/ssh/sshd_config

regexp: '^PermitRootLogin'

line: 'PermitRootLogin no'

backup: yes

notify: restart ssh

```
AWS [Alt+S] Search United States (N. Virginia) Account ID: 3111-4154-2113 Adrina Colaco
```

```
GNU nano 7.2 main.yml *
---
- name: Update apt cache
  apt:
    update_cache: yes

- name: Upgrade all packages
  apt:
    upgrade: dist
    register: upgrade_result

- name: Install common packages
  apt:
    name:
      - git
      - curl
      - ufw
      - build-essential
    state: present
    update_cache: yes

- name: Ensure UFW allows SSH
  ufw:
    rule: allow
    port: 22
    proto: tcp

- name: Ensure UFW allows HTTP
  ufw:
    rule: allow
    port: 80
    proto: tcp

- name: Enable UFW
  ufw:
    state: enabled

- name: Disable SSH password authentication
  lineinfile:
    path: /etc/ssh/sshd_config
    regexp: "PasswordAuthentication"
    line: 'PasswordAuthentication no'
    backup: yes
    notify: restart ssh

- name: Disable root login
  lineinfile:
    path: /etc/ssh/sshd_config
    regexp: "PermitRootLogin"
    line: 'PermitRootLogin no'
    backup: yes
    notify: restart ssh

Help Write Out Where Is Cut Execute Location Undo Set Mark To Bracket Previous Back
Exit Read File Replace Paste Justify Go To Line Redo Copy Where Was Next Forward
```

i-082f0d8b3ad04ba3b (LinuxAdrina)
PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

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ansible/roles/common/handlers/main.yml:

- name: restart ssh

Ansible.builtin.service:

name: ssh

state: restarted

```
GNU nano 7.2 main.yml
---
- name: restart ssh
  ansible.builtin.service:
    name: ssh
    state: restarted
```

Role: nginx

ansible/roles/nginx/tasks/main.yml:

- name: Install Nginx

apt:

name: nginx

state: present

update_cache: yes

- name: Copy Nginx site configuration

template:

src: epicbook.conf.j2

dest: /etc/nginx/sites-available/epicbook.conf

notify: reload nginx

- name: Enable site (create symlink)

file:

src: /etc/nginx/sites-available/epicbook.conf

dest: /etc/nginx/sites-enabled/epicbook.conf

state: link

- name: Remove default site (optional)

file:

path: /etc/nginx/sites-enabled/default

state: absent

```
GNU nano 7.2 main.yml
---
- name: Install Nginx
  apt:
    name: nginx
    state: present
    update_cache: yes

- name: Copy Nginx site configuration
  template:
    src: epicbook.conf.j2
    dest: /etc/nginx/sites-available/epicbook.conf
    notify: reload nginx

- name: Enable site (create symlink)
  file:
    src: /etc/nginx/sites-available/epicbook.conf
    dest: /etc/nginx/sites-enabled/epicbook.conf
    state: link

- name: Remove default site (optional)
  file:
    path: /etc/nginx/sites-enabled/default
    state: absent
```

ansible/roles/nginx/handlers/main.yml:

```
---
- name: reload nginx
  service:
    name: nginx
    state: reloaded
```

```
GNU nano 7.2 main.yml
---
- name: reload nginx
  service:
    name: nginx
    state: reloaded
```

ansible/roles/nginx/templates/epicbook.conf.j2:

```
server {
    listen 80;
    server_name _;
    root {{ epicbook_dest }};
    index index.html;

    location / {
```

```
    try_files $uri $uri/ =404;
}

error_log /var/log/nginx/epicbook_error.log;
access_log /var/log/nginx/epicbook_access.log;
}
```

```
GNU nano 7.2 epicbook.conf.j2
server {
    listen 80;
    server_name _;
    root {{ epicbook_dest }};
    index index.html;

    location / {
        try_files $uri $uri/ =404;
    }

    error_log /var/log/nginx/epicbook_error.log;
    access_log /var/log/nginx/epicbook_access.log;
}
```

Role: epicbook

ansible/roles/epicbook/tasks/main.yml:

- name: Ensure epicbook directory exists

file:

path: "{{ epicbook_dest }}"

state: directory

owner: www-data

group: www-data

mode: '0755'

- name: Clone the EpicBook repo

git:

repo: "{{ epicbook_repo_url }}"

dest: "{{ epicbook_dest }}"

version: "main" # adjust if branch different

- name: Set ownership of app content

file:

path: "{{ epicbook_dest }}"

recurse: yes

```
owner: www-data
group: www-data
```

- name: Reload Nginx to serve new content
- ```
ansible.builtin.service:
 name: nginx
 state: reloaded
```

```
GNU nano 7.2 main.yml
--
- name: Ensure epicbook directory exists
 file:
 path: "{{ epicbook_dest }}"
 state: directory
 owner: www-data
 group: www-data
 mode: '0755'

- name: Clone the EpicBook repo
 git:
 repo: "{{ epicbook_repo_url }}"
 dest: "{{ epicbook_dest }}"
 version: "main" # adjust if branch different

- name: Set ownership of app content
 file:
 path: "{{ epicbook_dest }}"
 recurse: yes
 owner: www-data
 group: www-data

- name: Reload Nginx to serve new content
 ansible.builtin.service:
 name: nginx
 state: reloaded
```

```
GNU nano 7.2 main.yml
- name: reload nginx
 ansible.builtin.service:
 name: nginx
 state: reloaded
```

You may add dependencies (node / build steps) if the repo requires building; but if it's static HTML / CSS then clone + reload is enough.

---

## Step 5: Run Ansible to Deploy

From the ansible/ folder:

```
ansible-playbook -i inventory.ini site.yml
```

```

ubuntu@ip-172-31-23-116:~/epicbook-prod/ansible$ ansible-playbook -i inventory.ini site.yml

PLAY [Baseline / Common configuration] *****

TASK [Gathering Facts] *****
ok: [54.242.56.32]

TASK [common : Update apt cache] *****
changed: [54.242.56.32]

TASK [common : Upgrade all packages] *****
ok: [54.242.56.32]

TASK [common : Install common packages] *****
ok: [54.242.56.32]

TASK [common : Ensure UFW allows SSH] *****
ok: [54.242.56.32]

TASK [common : Ensure UFW allows HTTP] *****
ok: [54.242.56.32]

TASK [common : Enable UFW] *****
ok: [54.242.56.32]

TASK [common : Disable SSH password authentication] *****
ok: [54.242.56.32]

TASK [common : Disable root login] *****
ERROR! The requested handler 'restart ssh' was not found in either the main handlers list nor in the listening handlers li
ubuntu@ip-172-31-23-116:~/epicbook-prod/ansible$ cd ..
ubuntu@ip-172-31-23-116:~/epicbook-prod$ cd ansible/^C

```

i-082f0d8b3ad04ba3b (LinuxAdrina)

PublicIPs: 3.84.114.45 PrivateIPs: 172.31.23.116

You should see:

- common role executing (updating packages, firewall, SSH hardening)
- nginx role installing & configuring Nginx with your template
- epicbook role cloning the repo and reloading Nginx
- Idempotency: re-run should show most “ok / unchanged” on repeated runs
- Test in browser: [http://<public\\_ip>](http://<public_ip>) should show your EpicBook app’s content

## Troubleshooting Tips & Verification

- Verify security group allows port 80 & 22
- Check `/var/www/epicbook/index.html` exists on the VM
- Tail Nginx logs if you get 502 or 404
- Re-run Ansible with `-v` or `-vvv` to debug failures
- Examine role order: ensure common runs before nginx before epicbook

LinkedIn Post1:

[https://www.linkedin.com/posts/adrina-colaco\\_terraform-ansible-devops-activity-7390094203793547264-n0lt?utm\\_source=share&utm\\_medium=member\\_desktop&rcm=ACoAAAPhxLwBOfaV\\_xhbVwfcuz8dVOvN7T9q0RM](https://www.linkedin.com/posts/adrina-colaco_terraform-ansible-devops-activity-7390094203793547264-n0lt?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhxLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM)

LinkedIn post2:

[https://www.linkedin.com/posts/adrina-colaco\\_terraform-ansible-azure-activity-7390099772621189120-CLep?utm\\_source=share&utm\\_medium=member\\_desktop&rcm=ACoAAAPhxLwBOfaV\\_xhbVwfcuz8dVOvN7T9q0RM](https://www.linkedin.com/posts/adrina-colaco_terraform-ansible-azure-activity-7390099772621189120-CLep?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhxLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM)

LinkedIn post3:

[https://www.linkedin.com/posts/adrina-colaco\\_terraform-ansible-aws-activity-7390102660827496448-yfie?utm\\_source=share&utm\\_medium=member\\_desktop&rcm=ACoAAAPhxLwBOfaV\\_xhbVwfcuz8dVOvN7T9q0RM](https://www.linkedin.com/posts/adrina-colaco_terraform-ansible-aws-activity-7390102660827496448-yfie?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhxLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM)

Blog : <https://www.linkedin.com/pulse/my-devops-journey-automating-infrastructure-terraform-adrina-colaco-zfa2c>

# The End

**P.S.** This document is part of the **FREE DevOps Micro Internship (DMI) Cohort** run by [Pravin Mishra](#). You may join the next FREE COHORT by filling this [Pre-registration](#) Form.